Process Characterization Report

A-Mab Drug Substance — Production Bioreactor (Step 3) — PCR-003

Process Development / Manufacturing Science & Technology (MSAT)

2026-09-30

Table of contents

Approvals 5
Revision history 6
Abbreviations 6

Executive summary **8**

1 Introduction **10**
1.1 Product and unit operation 10
1.2 Objectives 11
1.3 Regulatory and scientific basis 11
1.4 Changes in this revision 12

2 Prior knowledge and quality risk basis **14**
2.1 Platform and prior-product knowledge 14
2.2 Quality attributes in scope 15
2.3 Risk-based prioritization and parameter selection 16

3 Materials and methods **17**
3.1 Scale-down model and its qualification 17
3.2 Cell line, media and culture operation 18
3.3 Analytical methods 19
3.4 Sampling plan 20
3.5 Statistical methods 20

4 Study design **24**
4.1 Factors, ranges and the knowledge space 24
4.2 Screening design 25
4.3 Response-surface design 25
4.4 Univariate assessment 26
4.5 Osmolality study 26
4.6 Execution of the studies 26

5 Results **28**
5.1 Centre-point performance and reproducibility 28
5.2 Screening: factor effects 28
5.3 Response-surface models 30
5.3.1 Selected models and their adequacy 30
5.3.2 Residual patterns and influential runs 33
5.3.3 Effects of the parameters 36
5.3.4 What the augmentation changed 39
5.3.5 Near-equivalent models 40

5.4 Univariate assessment . . . 40
5.5 Osmolality study . . . 43
5.6 Mechanistic interpretation . . . 44

6 Design space **46**

7 Proven acceptable ranges **48**
7.1 Acceptance criteria . . . 48
7.2 Outputs against their criteria over the NOR . . . 50
7.3 PARs . . . 52
7.4 Univariate parameters and osmolality . . . 56

8 Process capability and robustness **57**

9 Parameter classification **59**

10 Contribution to the control strategy **61**

11 Discussion **63**

12 Conclusions **64**

13 Study events and discrepancies from the plan **65**
13.1 DEV-003-01, pCO2 probe drift during screening run 4 . . . 65
13.2 DEV-003-02, nutrient feed-1 under-delivery on a centre-point batch . . . 66
13.3 PD-003-01, osmolality study . . . 72
13.4 PD-003-02, augmentation of the response-surface design . . . 73
13.5 PD-003-03, model selection . . . 73
13.6 PD-003-04, models that do not meet the criterion on residuals and influential runs . . . 73
13.7 PD-003-05, duplicate univariate runs . . . 73
13.8 PD-003-06, setting of osmolality . . . 73
13.9 PD-003-07, osmolality not carried into the response-surface design . . . 74
13.10 PD-003-08, NOR not narrowed for acidic variants . . . 74
13.11 PD-003-09, viable cell density, viability and titre not in the record . . . 74
13.12 PD-003-10, univariate study started before the response-surface study . 74
13.13 PD-003-11, initial viable cell concentration not carried into a multivariate study . . . 75
13.14 PD-003-12, equipment and calibration status not recorded for the osmolality and augmentation studies . . . 75
13.15 PD-003-13, aggregate offset of the scale-down model accounted for, not investigated . . . 75

Appendix A. Screening design matrix and responses **77**

Appendix B. Response-surface design matrix and responses **79**

Appendix C. Full effect and coefficient tables **81**

Appendix D. Analytical methods summary **86**

References **87**

Field	Value
Document ID	PCR-003
Document class	Process Characterization Report
Title	Process Characterization Report: Production Bioreactor (Step 3)
Product	A-Mab
Version	2.0
Effective date	2026-09-30
Supersedes	PCR-003 Version 1.0 (2026-07-24)
Status	Approved (synthetic)
Document owner	Manufacturing Science & Technology (MSAT)
Sponsor	Novacyte Biologics
Development site	Cambridge, MA
Commercial drug substance site	Grafton, WI

SYNTHETIC DOCUMENT — NOT A REAL RECORD. Illustrative content generated from a seeded model of the *A-Mab* case study (CMC Biotech Working Group, 2009). Novacyte Biologics and all sites, SOP/AMV numbers and signatories are fictional. For NLP corpus development only; not for any regulated use.

Approvals

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Synthetic record. Signatures are not applied.

Revision history

Version	Effective date	Description of change
1.0	2026-07-24	Initial issue
2.0	2026-09-30	Reissued under INV-001 and INV-002: a priori intermediate acceptance criteria, re-analysis of the studies, execution dates recorded; changes listed in the document (see INV-001)

Abbreviations

Abbreviation	Meaning
2-AB	2-aminobenzamide
ADCC	antibody-dependent cellular cytotoxicity
Afuc.	afucosylation (table header)
AICc	Akaike’s information criterion with the small-sample correction
AMD	analytical method decision
AMV	analytical method validation
BLA	Biologics License Application
CDC	complement-dependent cytotoxicity
CHO	Chinese hamster ovary
CI	confidence interval
Cpk	process capability index
CPP	critical process parameter
CQA	critical quality attribute
DoE	design of experiments
DS	drug substance
ELISA	enzyme-linked immunosorbent assay
FcγRIIIa	Fc gamma receptor IIIa
Gal.	galactosylation (table header)
GDP	guanosine diphosphate
GPP	general process parameter
H	high (criticality level)
HCP	host cell protein
HILIC	hydrophilic interaction liquid chromatography
HMW	high molecular weight species (aggregate)
IAC	intermediate acceptance criterion
icIEF	imaged capillary isoelectric focusing
IgG1	immunoglobulin G, subclass 1
KPP	key process parameter
LMW	low molecular weight species (fragments)
LOQ	limit of quantitation

Abbreviation	Meaning
M-H	moderate to high (criticality level)
Mfg	manufacturing scale (table header)
MSAT	Manufacturing Science and Technology
n.t.	not tested
NOR	normal operating range
nrCE-SDS	non-reduced capillary electrophoresis with sodium dodecyl sulfate
PAR	proven acceptable range
pCO2	CO2 partial pressure, the partial pressure of dissolved carbon dioxide
PPQ	process performance qualification
PRESS	prediction error sum of squares
QA	Quality Assurance
qPCR	quantitative polymerase chain reaction
RMSEP	root mean square error of prediction
SD	standard deviation
SDM	scale-down model
SEC	size exclusion chromatography
SOP	standard operating procedure
UDP	uridine diphosphate
UF/DF	ultrafiltration and diafiltration
UPLC	ultra-performance liquid chromatography
VH	very high (criticality level)
VL	very low (criticality level)
WC-CPP	well-controlled critical process parameter

Executive summary

This report gives the results of the Stage 1 characterization of the production bioreactor, Step 3 of the A-Mab drug substance process. The production bioreactor is a fed-batch culture with a working volume of 15,000 L at commercial scale, and it forms the glycan, charge variant and aggregate quality attributes of A-Mab. The studies were run in a qualified scale-down model under the Process Characterization Plan PCP-003 and the Process Characterization Master Plan PCMP-001, following ICH Q8(R2), ICH Q11 and the A-Mab case study (International Council for Harmonisation 2009, 2012; CMC Biotech Working Group 2009). This is Version 2.0 of the report. It replaces Version 1.0 after the QA investigations INV-001, on the acceptance criteria, and INV-002, on the adequacy of the response-surface models (Section 1.4).

The scale-down model was compared with manufacturing scale in three runs at the set-points against five manufacturing-scale batches. A screening design of 19 runs varied the five parameters assigned to multivariate study, and a face-centred response-surface design of 28 runs then varied four of them. Under INV-002, 16 runs were added to the response-surface design as a second block, after the results of the planned runs were known. A univariate study varied the other four parameters one at a time. An osmolality study, made during the Version 2.0 re-analysis, varied osmolality with the other parameters at the combination of NOR settings least favourable for acidic variants.

The response-surface model of each of the five modelled attributes was selected on all 44 runs, by the model selection that INV-002 adopted as its corrective action CA3. The scale comes from the Box-Cox profile, and the model is the one with the lowest AICc among all hierarchical submodels of the full quadratic. Every selected model meets the first three adequacy criteria of PCMP-001 §7.2, a significant model F-test, no significant lack of fit and predicted R^2 within 0.20 of R^2. For galactosylation and acidic variants the absolute residuals rise with the predicted value, so these two models do not meet the fourth criterion, on residuals and influential runs. They were used with that trend sized against their criteria (PD-003-04). With each run flagged by its Cook's distance deleted in turn, every attribute kept its verdict over the normal operating range (NOR), and every proven acceptable range (PAR) still contained the NOR.

Each attribute was judged on the 95 % prediction interval of a single run, at manufacturing scale, at every combination of NOR low, set-point and NOR high. Afucosylation, galactosylation, high mannose and aggregate meet their intermediate acceptance criteria. Acidic variants do not. At nine of the 81 combinations, all with pCO2 at its NOR high, the interval reaches below the lower limit of 18.0 %. At the least favourable combination the predicted median is 18.4 %, a margin above the limit smaller than the bias of the model at the set-point, and about 17.7 % with the initial viable cell concentration at its NOR low. The NOR was not narrowed. The risk was accepted as the

producer’s. This report recommends a combined alert in the control system, which fires when culture pH is above 6.79, pCO2 is above 101 mmHg and culture temperature is below 35.39 °C at the same time. It also recommends an action limit for acidic variants at harvest of 18.72 %, and the trending of acidic variants against the initial viable cell concentration in production (Section 10).

All 20 PARs computed with the other parameters varying across their NORs contain the NOR, and 18 of them span the whole range studied. That every PAR contains the NOR rests on the 16 augmentation runs and on the model selection. On the 28 planned runs alone, the PARs of culture pH, culture temperature and pCO2 for acidic variants do not contain it. The design space covers 73.7 % of the region studied, and its boundary is set by the lower limit of acidic variants.

Culture pH, culture temperature, pCO2 and culture duration were classified as well-controlled critical process parameters (WC-CPPs). Osmolality was classified as a critical process parameter (CPP), because no PAR from a model shows a wide margin between its acceptable range and its NOR. The initial viable cell concentration, basal medium concentration and nutrient feed-1 volume were classified as key process parameters (KPPs), and dissolved oxygen as a general process parameter (GPP). In the integrated model of the train the capability index of acidic variants in the drug substance is a Cpk of 1.05, between the minimum of 1.00 and the target of 1.33, and every other attribute that the step sets has a Cpk above the target. That model simulates manufacturing scale directly and carries no effect of the initial viable cell concentration on acidic variants (Section 8).

The record of the step holds two study events and 13 discrepancies from the plan (Section 13). The pCO2 probe drift in one screening run and the feed under-delivery to one centre-point batch (DEV-003-01 and DEV-003-02) left their runs in the analysis. Seven of the discrepancies are requirements of PCP-003 that the Version 2.0 re-analysis found unmet, among them the multivariate study that the univariate effect of the initial viable cell concentration on acidic variants calls for (PD-003-11).

1 Introduction

1.1 Product and unit operation

A-Mab is a humanized IgG1 monoclonal antibody whose main mechanism of action is antibody-dependent cellular cytotoxicity (ADCC). It is produced by a recombinant Chinese hamster ovary (CHO) cell line and purified by the platform train of Protein A capture, low-pH viral inactivation, cation exchange, anion exchange, small-virus retentive filtration and ultrafiltration and diafiltration (UF/DF) (CMC Biotech Working Group 2009).

The production bioreactor is Step 3 of the drug substance process. It is a fed-batch culture with a working volume of 15,000 L at commercial scale. The cells secrete A-Mab into the medium over a culture of 17 days at the set-point, and the culture fluid passes to harvest and clarification (Step 4) at the end of the run. The production bioreactor is the only step of the A-Mab drug substance process at which product quality attributes are formed. Cation and anion exchange separate by net charge, and core fucose, galactose and mannose carry none, so no purification step retains one neutral glycoform in preference to another (PCP-003 §4.1). The neutral glycoforms of the drug substance are therefore those of the harvest. They are set mainly in the Golgi before secretion, and glycosidases released by lysed cells can trim them further in the culture fluid. Acidic variants form in the cell and in the culture fluid, mainly as asparagine deamidates, and aggregate forms as antibody molecules associate in the culture fluid. Both rise with the time the antibody spends at culture temperature. Host cell protein and DNA enter the culture fluid when cells die and lyse, so their load at harvest rises as viability falls late in the culture.

This report gives the results of the studies planned in PCP-003. It applies the strategy, the statistical methods and the acceptance criteria of PCMP-001 Version 2.0, and it rolls up into the Process Characterization Master Report PCMR-001. The risk assessment RA-001 set the scope of the studies. The characterization supports the transfer of the process from the development site in Cambridge, MA to the commercial drug substance site in Grafton, WI under the Process Transfer Plan PTP-001. Table 1.1 lists these documents.

Table 1.1: Documents related to this report.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope

Document ID	Title	Relationship
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCP-003	Process Characterization Plan (Production Bioreactor (Step 3))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into
INV-001	Investigation Report (Acceptance criteria for proven-acceptable-range analysis set after data review)	Reason for this revision
INV-002	Investigation Report (Response-surface models below the model-adequacy criteria of PCMP-001 Version 2.0)	Model adequacy of the response-surface designs

1.2 Objectives

PCP-003 §2 sets the following objectives, and Section 12 states the result against each.

1. To quantify the relationship between each studied parameter and each of the five modelled responses over the ranges studied, and to identify the interactions between parameters.
2. To define the multivariate region of the response-surface parameters over which every modelled attribute is predicted to meet its intermediate acceptance criterion.
3. To establish a PAR for each parameter against each attribute whose fitted model contains a significant term in that parameter, and to confirm that the NOR lies inside that range.
4. To provide the demonstrated effects and the operating risk from which each parameter is classified under the parameter classification procedure SOP-4001.
5. To provide the process understanding and the capability estimate that justify the Stage 2 operating ranges and the contribution of this step to the control strategy.

1.3 Regulatory and scientific basis

The studies are Stage 1 process design studies in the lifecycle approach to process validation (U.S. Food and Drug Administration 2011). They follow the enhanced approach of ICH Q8(R2) and ICH Q11. Prior knowledge and risk assessment identify the parameters that may affect an attribute, and designed studies then measure and bound those effects (International Council for Harmonisation 2009, 2012). ICH Q8(R2) defines the design space and the proven acceptable range, and it treats the criticality of an attribute as a continuum. The risk tools follow ICH Q9 (International Council for Harmonisation 2023). The A-Mab case study supplies the process model,

the framework of quality attributes and the decision logic of parameter classification (CMC Biotech Working Group 2009). The designs and their analysis use response-surface methods (Montgomery 2017). The model selection uses the Box-Cox transformation and Akaike's information criterion with the small-sample correction (AICc), and the augmentation of the response-surface design is D-optimal (Box and Cox 1964; Hurvich and Tsai 1989; Burnham and Anderson 2002; Fedorov 1972).

1.4 Changes in this revision

Version 1.0 of this report was approved on 2026-07-24. QA opened the investigation INV-001 on 2026-08-03 after its review of the master report PCMR-001 Version 1.0 for BLA readiness, and closed it on 2026-09-18 with the classification major. The investigation found that the criteria used to judge each step in the analysis of the proven acceptable ranges were in no approved protocol and had been set after the data were seen (INV-001 finding 1). For this step, the Version 1.0 limits of the glycans and acidic variants were the mean plus 2.5 standard deviations of the capability data they were then used to judge (finding 3). The Version 1.0 aggregate limit was the drug substance limit, back-calculated through the nominal downstream clearance of this campaign and divided by a margin of 6.8 (finding 2). Acidic variants carry the two-sided drug substance criterion of 18–40 %, and Version 1.0 assessed their capability against its upper end alone and judged their ranges against an upper limit alone, ≤ 25.8 % (finding 4). The records gave no execution dates (finding 5). The second investigation, INV-002, was opened on 2026-08-31 and closed on 2026-09-28 with the classification minor. It found that the full quadratic models of high mannose and acidic variants on the planned runs did not meet the limit on overfitting of PCMP-001 §7.2.

This revision makes the following changes.

1. Each output is judged against an intermediate acceptance criterion that PCMP-001 Version 2.0 derives a priori from the drug substance criterion and the platform worst case of every downstream step (Table 1.2). The Version 1.0 limits were applied at analysis and appeared in no approved plan. Every upper limit in force is higher than the Version 1.0 limit, and acidic variants now carry a lower limit of 18.0 %. Verification of the platform worst cases tightened the aggregate criterion and then widened it, to 6.30 % against 9.30 % a priori (Section 7.1).
2. Acidic variants are judged against both limits of their criterion, and their capability is assessed against both (Section 8).
3. The scale-down model is qualified for each output by its offset from manufacturing scale, and every result judged against a criterion is stated at manufacturing scale (Section 3.1).
4. Every response-surface output is analysed by model selection on all runs of the design, with a block term (PD-003-03). The design was augmented by 16 runs under INV-002 (PD-003-02).
5. The osmolality study S-BRX-OSM was made during the re-analysis (PD-003-01).
6. The execution window of every study is recorded, and for every study of PCP-003 the calibration status of the pCO2 probe (Section 4.6).

7. The results for host cell protein and DNA in the clarified harvest, which PCP-003 requires and Version 1.0 did not report, are given for information (Section 3.4).
8. The description of the control of culture pH is corrected. Version 1.0 gave it as control by carbon dioxide and base addition, but carbon dioxide cannot set the pH where pCO2 is a factor of the designs (Section 3.2).
9. Version 1.0 recorded the events of execution as deviations. A process characterization study is not GMP manufacture, so this report records them as study events, and every departure from PCP-003 as a discrepancy from the plan. The departures that the Version 2.0 re-analysis found are PD-003-07 to PD-003-13 (Section 13).

The Version 1.0 analysis judged the full quadratic models of the planned runs against the Version 1.0 limits. Judged so, 13 of the 20 NOR-propagated PARs do not contain the NOR, and each of the five attributes has at least one such PAR. Judged against the criteria in force, the same models give the PAR of pCO2 for acidic variants as 40–115 mmHg, short of the NOR high, as INV-001 reports. The PARs of this report come from the selected models on all runs (Section 5.3.4). The lower limit of acidic variants, which Version 1.0 did not apply, is the one criterion not met across the NOR (Section 7.2). The analytical method decision AMD-001 of 2026-03-20, which moved the measurement of fragments to non-reduced CE-SDS, changes no method of this report (Table 3.3).

Table 1.2: Acceptance criteria for the step outputs in Version 1.0 and Version 2.0, production bioreactor.

Attribute	Version 1.0[a)]		Version 2.0	
	Criterion	Rule	Criterion	Basis
Acidic variants (%)	≤ 25.8	R2	18.0–39.0	Carried back
Afucosylation (%)	2.00–9.43	R2	2.00–13.0	Drug substance
Aggregates, HMW (%)	≤ 1.86	R1, margin 6.8	≤ 6.30	Carried back, tightened[b)] and widened[c)] from ≤ 9.30
Galactosylation (%)	10.0–30.1	R2	10.0–40.0	Drug substance
High mannose (%)	3.00–7.28	R2	3.00–10.0	Drug substance

[a)] Applied in the Version 1.0 reports at analysis. No approved Version 1.0 plan contained these criteria (INV-001 finding 1).
[b)] Tightened on verification: a downstream step did worse within its NOR than the platform worst case.
[c)] Widened on verification: a downstream step did better than the platform worst case at the one-sided 95 % prediction bound of a single batch, on a model that meets the first three adequacy criteria of PCMP-001 §7.2.
[R1] Version 1.0 rule: nominal back-calculation / margin.
[R2] Version 1.0 rule: mean + 2.5 sd of capability.

2 Prior knowledge and quality risk basis

2.1 Platform and prior-product knowledge

The platform on which A-Mab is produced was used for the humanized IgG1 antibodies X-Mab, Y-Mab and Z-Mab and for the A-Mab clinical and engineering campaigns (CMC Biotech Working Group 2009). Cell line construction, basal medium, feed strategy and the control of pH, temperature and dissolved oxygen are common to these products. The directions of the effects seen on the platform are carried into this study as expectations. Their sizes are not, and the fitted models of this study give them.

The glycans are built in the Golgi by enzymes that draw on nucleotide sugar donors. Core fucose is added by a fucosyltransferase from GDP-fucose. The fraction of antibody secreted without core fucose therefore rises when the activity of that enzyme falls, when the donor is depleted or when the glycan spends less time in the Golgi. Terminal galactose is added by a galactosyltransferase from UDP-galactose. The donor and the manganese cofactor of the enzyme are depleted late in a fed-batch culture, so galactosylation is expected to fall as the culture ages. High mannose glycans remain when processing stops early, so their level rises when transit through the Golgi is fast relative to the processing enzymes or when those enzymes are inhibited. The harvested pool contains everything secreted during the run, so a longer culture weights it toward antibody made late, under depleted conditions.

A higher culture pH raises the pH inside the cell and in the Golgi. The glycosyltransferases and the mannosidases each work fastest at a different pH, so a shift changes their rates by different amounts, and its direction on each glycan is read from the data. A higher culture temperature raises the growth rate and the rate of every enzymatic and chemical reaction. Dissolved carbon dioxide forms bicarbonate and lowers the pH in the cytosol, so a higher pCO2 is expected to move each glycan in the direction opposite to a higher culture pH. A higher osmolality draws water out of the cell, slows its growth and changes the size of the nucleotide sugar pools. These four parameters act on the same enzymes and donor pools, so the effect of one on a glycan is expected to depend on the levels of the others (PCP-003 §4.1).

Aggregate forms after secretion, while the antibody sits in the culture fluid at culture temperature. A longer culture raises it through the longer time at temperature. A culture pH closer to the isoelectric point raises it, because the net charge of the antibody falls and the repulsion between molecules weakens. Acidic variants are mainly products of asparagine deamidation, together with sialylated glycans and glycated lysines. Deamidation speeds up at a higher temperature and a higher pH. Sialylation rises with the activity of the sialyltransferases inside the cell, and glycation rises with the residual glucose of the medium. These routes do not respond to a parameter in

the same direction, so the net direction of each parameter on acidic variants is read from the data.

Dissolved oxygen supplies the oxygen for oxidative metabolism, and the culture is not expected to be limited by oxygen at either end of the range studied, so neither growth nor the glycans are expected to move across it. A higher initial viable cell concentration brings the culture to its peak viable cell density earlier and leaves more days in the productive phase, which raises the titre. A higher basal medium concentration and a larger nutrient feed-1 volume raise the supply of glucose, amino acids and nucleotide sugar precursors, which raises growth and titre. PCP-003 §4.1 expects none of these four parameters to move a glycan attribute at a fixed culture age within the ranges studied. The record holds no metabolite data with which to check the supply directly.

2.2 Quality attributes in scope

Table 2.1 lists the seven quality attributes that the drug substance CQA register assigns to the production bioreactor. Criticality was assigned in RA-001 with Tool #1 of the A-Mab case study, whose score is the product of the impact of the attribute and the uncertainty in that impact (CMC Biotech Working Group 2009). The levels run from very low (VL) to very high (VH), and the attributes of high (H) and very high criticality are called critical. For this step the critical attributes are afucosylation, galactosylation, high mannose and aggregate. Host cell protein has moderate to high criticality (M-H). Acidic variants and residual DNA have very low criticality, and their drug substance criteria still apply (PCMP-001 §3).

Table 2.1: Quality attributes that the drug substance CQA register assigns to the production bioreactor, with their drug substance criteria and criticality.

CQA	Category	Acceptance	Criticality	Tool #1
Afucosylation	glycosylation	2–13 % of glycans	H	60
Galactosylation (total %Gal)	glycosylation	10–40 % (G1+G2 equiv.)	H	48
High Mannose	glycosylation	3–10 % of glycans	VH	80
Aggregates (HMW)	size variant	≤ 5 % HMW (SEC)	H	60
Acidic charge variants (deamidation)	charge variant	18–40 % (CEX/icIEF)	VL	4
Host Cell Protein (HCP)	process impurity	≤ 100 ng/mg	M-H	36
Residual DNA	process impurity	≤ 0.001 ng/dose	VL	6

Antibody without core fucose binds the Fc gamma receptor IIIa (FcγRIIIa) on the effector cell more strongly, which raises ADCC, so afucosylation is linked to efficacy. In the A-Mab studies complement-dependent cytotoxicity (CDC) rose with the galactose content, while ADCC did not change (CMC Biotech Working Group 2009). High mannose species are cleared from the circulation faster than the fully processed glycoforms, which gives high mannose the highest Tool #1 score of the step. Aggregate is scored for its immunogenic potential.

The five modelled responses of the designs are afucosylation, galactosylation, high mannose, acidic variants and aggregate. Host cell protein and residual DNA were measured on the clarified harvest for information. Neither has an intermediate acceptance criterion at this step. Host cell protein is first judged in the Protein A pool. Residual DNA is judged on the drug substance only, because the platform history gives no worst-case DNA clearance for the steps after Protein A (PCMP-001 §7.3). Its criterion, ≤ 0.001 ng per dose, is the acceptable range that the A-Mab case study claims, which is met through the clearance of the purification train (CMC Biotech Working Group 2009, 47). A dose is 10 mg/kg, given weekly in A-Mab, for a reference adult of 70 kg, an assumption of this campaign. The CQA register assigns fragments to harvest, and fragments have no intermediate acceptance criterion at this step, so they are not an attribute of these studies.

2.3 Risk-based prioritization and parameter selection

The risk assessment RA-001 ranked every parameter of the step before characterization began, on its potential impact on a quality attribute and its potential to interact with other parameters (International Council for Harmonisation 2023). Parameters ranked high, or with a likely interaction, were assigned to multivariate study, and the others to univariate study (PCMP-001 §4). For the production bioreactor five of the nine parameters were assigned to multivariate study, namely culture pH, culture temperature, CO2 partial pressure (pCO2), osmolality and culture duration. Culture pH, pCO2 and osmolality change the pH inside the cell or the nucleotide sugar pools, culture temperature changes the rate of growth and of the Golgi enzymes, and culture duration weights the harvested pool toward antibody made late in the culture. The effect of one is therefore expected to depend on the levels of the others, and only a design that varies them together can estimate that dependence. Dissolved oxygen, initial viable cell concentration, basal medium concentration and nutrient feed-1 volume were assigned to univariate study, because none of them is expected to move a glycan attribute at a fixed culture age (Section 2.1).

The range studied of every parameter extends beyond its NOR at both ends, and it is between 1.5 and 2.5 times as wide as the NOR. The wider range shows how each attribute responds beyond the edges of routine operation, and it sets the knowledge space within which a PAR can be claimed. No PAR is claimed beyond the range studied.

3 Materials and methods

3.1 Scale-down model and its qualification

The studies were run in bench-scale stirred-tank bioreactors of the same design class as the commercial production vessel, with the same control scheme for pH, temperature, dissolved oxygen and nutrient addition. The scale-down model was operated under the bioreactor procedure SOP-2003. Parameters that do not depend on scale were run at the commercial set-points. Parameters that depend on scale, such as the agitation rate, the gas flow rates and the working volume, were set so that the small scale matched the performance of the full scale (CMC Biotech Working Group 2009). pCO2 is the exception. A large vessel strips carbon dioxide less efficiently than a bench-scale vessel, so pCO2 was controlled to the level of each run and studied as a factor, over a range that spans the levels the commercial vessel reaches (PCP-003 §5.1). The pCO2 of each run was measured by the probe of the scale-down bioreactor system, EQ-BRX-205.

The qualification study S-BRX-SDM (2025-08-11 to 2025-09-05) followed the scale-down model procedure SOP-1001. It compared three runs of the scale-down model at the set-points with five manufacturing-scale batches of the A-Mab engineering and clinical campaigns. A scale-down model need not reproduce the manufacturing mean, provided the difference is known well enough that no conclusion depends on its exact size (PCMP-001 §7.1). The offset of an output is the mean of the scale-down runs minus the mean of the manufacturing batches, with its two-sided 90 % confidence interval. For an output modelled on the log scale (Section 3.5) the offset is the ratio of the geometric means, and a scale-down value is divided by it. Every result of the scale-down studies that is judged against a criterion has the offset applied, so ranges and verdicts are stated at manufacturing scale. The model is qualified for an output when the verdict over the NOR, whether the output meets its criterion at every combination of NOR settings, is the same with the offset at either end of its interval.

Table 3.1 gives the offsets, as differences for the two outputs modelled on the raw scale and as ratios for the three modelled on the log scale. The scale-down model is qualified for every output with a criterion. Acidic variants fail their criterion over the NOR with the offset at its estimate and at either end of its interval, so that verdict does not depend on the offset either. Only for aggregate does the interval exclude a ratio of 1, and the scale-down model gives about 8 % less aggregate than manufacturing scale. That difference is accounted for through the offset and was not investigated before the characterization runs (PD-003-13). Its cause is not known.

The qualification runs ended on 2025-09-05, before the screening study started on 2025-09-08. PCP-003 Version 1.0 compared the means of the two scales by a test of

no difference at that point, so the difference of the means was known before the first characterization run. The confidence intervals of Table 3.1 and the decision whether the model is qualified for each output were computed at analysis, as PCMP-001 §7.1 allows. The viable cell density, the viability at harvest and the titre, which PCP-003 §5.1 also lists among the outputs compared, are not in the record of the qualification or of any characterization run (PD-003-09).

The centre points of the screening and response-surface designs were run at the same set-points as the qualification runs. They therefore carry the variation between runs of the scale-down model together with the variation of the analytical methods, and they give the pure error of the lack-of-fit test. The offset was estimated at the set-points only, and its use across the range studied is an assumption of PCP-003 §12.

Table 3.1: Offsets of the scale-down models (scale-down minus manufacturing scale) and the qualification of each model for each output, production bioreactor.

Output	Runs		Mean		Offset		Met over NOR (low/est./high)	Qualified
	SDM	Mfg	SDM	Mfg	Estimate	90 % CI		
Afucosylation (%)	3	5	7.54	7.92	−0.38	−0.88 to 0.11	Yes / Yes / Yes	Yes
Galactosylation (%)			26.13	26.37	−0.24	−2.33 to 1.85	Yes / Yes / Yes	Yes
High mannose (%)			5.98[a)]	6.44[a)]	×0.928[a)]	×0.852 to ×1.011	Yes / Yes / Yes	Yes
Acidic variants (%)			21.27[a)]	21.85[a)]	×0.973[a)]	×0.896 to ×1.057	No / No / No	Yes
Aggregates, HMW (%)			1.43[a)]	1.56[a)]	×0.917[a)]	×0.873 to ×0.963	Yes / Yes / Yes	Yes

[a)] Modelled on the log scale: geometric means, and the offset as the ratio of the scale-down to the manufacturing geometric mean. The manufacturing-scale value is the scale-down value divided by the ratio. SDM, scale-down model. Mfg, manufacturing scale. Offset, scale-down mean minus manufacturing mean. Met over NOR: the criterion is met at every combination of NOR low, set-point and NOR high (the NOR in force), with the offset at the low end of its interval, at its estimate (est.) and at the high end.

3.2 Cell line, media and culture operation

Cells were thawed from the working cell bank and expanded through the seed train to the initial viable cell concentration of the run. The production bioreactor was operated in fed-batch mode under SOP-2003, with a single nutrient feed added on the platform schedule. Culture pH was held by base addition against the lactate that the culture produces, which lowers it. pCO2 was held at the level of each run by the fraction of carbon dioxide in the sparge gas, temperature by the vessel jacket and dissolved oxygen by sparging and agitation. This corrects the description of Version 1.0, which gave culture pH as controlled by carbon dioxide and base addition. Basal medium and feed were prepared under the media and feed procedure SOP-2004. Each factor of a design was held at the level the design gives for the run, every other parameter at its set-point, and the culture was harvested at the culture duration of the run.

Osmolality was set by sodium chloride added with each feed to the target osmolality, with pCO2 held at its level by the sparge. PCP-003 does not state how osmolality is brought to its design level in a run in which pCO2 also varies (PD-003-06). The osmolality study used the same procedure, so the designs estimate the effects of osmolality and pCO2 separately. In a commercial vessel more base is added to hold the pH against a higher pCO2, which raises the osmolality as bicarbonate and sodium accumulate, and Section 7.3 states what follows for the PAR of pCO2.

Inputs controlled to a platform specification, such as the medium and feed lots, the gases and the base, were tested for identity and quality on receipt and were not characterized. A mass reconciliation after the run detected the feed under-delivery of DEV-003-02.

3.3 Analytical methods

The released N-glycan map by 2-AB HILIC-UPLC measured afucosylation, galactosylation and high mannose from one preparation, so the three glycan results of a run are not independent measurements. Size exclusion chromatography (SEC) measured aggregate as high molecular weight species (HMW). Imaged capillary isoelectric focusing (icIEF) measured acidic variants. The host cell protein ELISA and the residual DNA qPCR measured the two impurities in the clarified harvest. Table 3.2 lists the procedures and method validations, Table 3.3 the method for each size variant, and Appendix D the precision of each method. Each method was used within its validated range. Method precision contributes to the variation between centre points, and so to the pure error, and it sets a floor under the smallest effect the study can resolve.

Table 3.2: Procedures and method validations referenced in this document.

Type	Reference	Title
Standard operating procedure	SOP-1001	Qualification of Scale-Down Models
	SOP-1002	Design, Execution and Statistical Analysis of DoE Studies
	SOP-2003	Operation of the Production Bioreactor (Fed-Batch)
	SOP-2004	Cell-Culture Media and Feed Preparation
	SOP-3010	Released N-Glycan Mapping by 2-AB HILIC-UPLC
	SOP-3011	Size-Variant Analysis by SEC-HPLC
	SOP-3012	Host-Cell-Protein Quantitation by ELISA
	SOP-3013	Charge-Variant Analysis by icIEF
	SOP-3014	Residual Host-Cell DNA by qPCR
	SOP-4001	Process-Parameter Classification and Control-Strategy Definition
Analytical method validation	AMV-3010	N-Glycan Map (2-AB HILIC-UPLC)
	AMV-3011	Size-Variants (SEC-HPLC)
	AMV-3012	Host-Cell Protein ELISA
	AMV-3013	Charge Variants (icIEF)
	AMV-3014	Residual DNA (qPCR)

Table 3.3: Methods used for size variants in the designed studies, production bioreactor.

Attribute	Method used in study	
	Screening	Response surface
Aggregates, HMW	SEC	
Fragments, LMW	not in scope[a)]	

[a)] Fragments were not an attribute of this study.

3.4 Sampling plan

Each run was sampled at harvest, and the harvest sample was assayed for the five modelled responses. The clarified harvest was assayed for host cell protein and residual DNA in the studies that Section 4.6 lists. Across the 63 runs assayed, host cell protein in the clarified harvest ranged from 228,000 to 1,320,000 ng/mg and DNA from 64,000 to 520,000 pg/mg. The step carries no criterion for either. The culture was monitored through each run for viable cell density, viability and the principal metabolites under SOP-2003, and these records are not in the study record (PD-003-09).

Each run carries the number it has in the standard order of its design, the “Run” column of the design tables in Appendices A and B, and this report cites runs by that number. The runs were made in randomized order. The three centre points of the screening design were made at positions 11, 15 and 19 of its 19 runs, and the four centre points of the response-surface design at positions 3, 4, 8 and 25 of its 28 runs. Of the augmentation runs, five repeat a setting of the planned design (runs 29, 33, 35, 41 and 44 repeat runs 2, 5, 7, 11 and 16). They lie in the other block, so the analysis does not count them as pure error, and the pure error of the models fitted to all runs comes from the centre points of the response-surface design.

3.5 Statistical methods

The designs were generated, randomized and analysed under SOP-1002, with the methods of PCMP-001 §6 (Montgomery 2017). Each factor was coded so that its set-point is 0 and the ends of its range studied are −1 and +1. For this step the set-point of every factor of the designs lies at the centre of its range studied, so the centre points of the designs are at the set-points.

The screening data were fitted by least squares to the five main effects and the ten two-factor interactions. An effect is twice the coded coefficient, the change in the response across the full range studied. The model has 16 terms with the intercept in 19 runs. It leaves one degree of freedom for lack of fit and two for pure error. With so few, R^2 overstates how well the model would predict new runs, so the screening analysis was read only to identify active factors and interactions. Its lack-of-fit test

compares the mean of the centre points with the mean of the factorial runs, which is the test for curvature.

PCP-003 prescribes the full quadratic model of each response-surface output, with 14 terms besides the intercept. The outputs were analysed instead by the model selection that INV-002 adopted on 2026-09-15 as its corrective action CA3, recorded as the discrepancy PD-003-03. It has three steps.

1. The scale. The Box-Cox profile of the full quadratic model gives the exponent λ that best makes the residuals normal with constant variance, with its 95 % interval (Box and Cox 1964). An output is modelled on the log scale where that interval contains 0, the log, and excludes 1, the raw scale. It is modelled on the raw scale otherwise.
2. The model. Among all 1,337 submodels of the full quadratic that keep hierarchy, the model with the lowest AICc is taken. A submodel keeps hierarchy when it contains an interaction only with both its main effects, and a square only with its main effect.
3. The check. The selected model is judged against the adequacy criteria of PCP-003 §7, listed below.

AICc estimates how well a model would predict new runs. It rewards a smaller residual and penalises every estimated term, and its small-sample correction raises the penalty when the runs are few for the number of terms, as in these designs (Hurvich and Tsai 1989; Burnham and Anderson 2002). A stepwise search changes one term at a time and stops when no single change lowers AICc, so it can stop where only a combination of changes would lower it further. On the data of this step a forward search would have stopped at a model with a higher AICc for acidic variants, and a backward search for aggregate.

The models were fitted to all 44 runs, the 28 planned runs and the 16 augmentation runs, with a block term coded −1 for the planned runs and +1 for the augmentation. A prediction is made with the block term at 0, the average of the two blocks.

Of the modelled attributes, high mannose, acidic variants and aggregate are modelled on the log scale. On that scale the model reads $\ln(y) = b0 + b1 \cdot x1 + \ldots + e$. Transformed back, $y = e^{b0} \cdot e^{b1 \cdot x1} \cdot \ldots \cdot e^{e}$, so a main effect multiplies the output by e^{2b} from the low to the high end of its range when the other factors are at their set-points, where the interaction terms vanish. The error between runs multiplies the output too, so its size in percentage points grows with the level of the output. For high mannose and acidic variants this is expected, because the precision of the glycan map and of icIEF is a percentage of the level (Appendix D). For aggregate the profile supports the log scale too, although the precision of SEC is an absolute amount. Least squares, and the tests and prediction intervals built on it, assume residuals that are independent and normally distributed with mean zero and the same variance over the whole range studied. On the raw scale, the residual standard deviation of these outputs in the upper half of the predicted values is 1.24 to 2.22 times that in the lower half. On the log scale the ratio is 0.81 to 1.88.

On the log scale the criterion stays in its own unit, percent. A prediction interval is calculated on the log scale and transformed back. It keeps its 95 % coverage, because

the probability that ln y lies between two values is the probability that y lies between their antilogs. It is asymmetric in percentage points, further above the predicted value than below it. At the least favourable NOR combination for acidic variants, for example, the interval runs 2.15 points below the predicted value and 2.43 points above it. The predicted value of a log-scale model is the median of the output, its geometric mean, and not its arithmetic mean. The prediction limits, which decide the verdict against a criterion, are not affected by that.

A selected model is adequate for prediction under PCP-003 §7 when all of the following hold. The first four restate the criteria of PCMP-001 §7.2.

a. The model F-test. The F-test of the model is significant at $\alpha = 0.05$.
b. The lack-of-fit test. Lack of fit against pure error is not significant at $\alpha = 0.05$.
c. The limit on overfitting. Predicted R^2 falls short of R^2 by no more than 0.20. Predicted R^2 comes from the prediction error sum of squares (PRESS).
d. The criterion on residuals and influential runs. A regression of the absolute residuals on the predicted values, and a regression of the residuals on the run order, each show no slope significant at $\alpha = 0.05$. No run has a Cook's distance above the median of the F distribution with p and $n - p$ degrees of freedom, the F median, where p is the number of parameters of the model and n the number of runs. A run with a Cook's distance above 4/n, which is 0.091 for the 44 runs of the models, is examined by deleting it and repeating the analysis. The augmentation runs continue the order of the planned runs.
e. The normality criterion. The normal quantile plot of the residuals shows no departure from normality.

Each run above 4/n was deleted in turn and the model selection repeated. Each attribute was then judged over the NOR again, and its NOR-propagated PARs and the design space were computed again. The models that do not meet the criterion on residuals and influential runs were used for the ranges and the design space, with each residual trend sized against the criterion of its output (PD-003-04).

Other models may have an AICc within 2 units of the lowest and fit the data almost as well (Burnham and Anderson 2002). They were computed for every output, and Section 5.3.5 states where they differ from the selected model.

Each output is judged against its intermediate acceptance criterion at manufacturing scale. A quality attribute is judged on the exact 95 % prediction interval of a single run at each of the 81 combinations of NOR low, set-point and NOR high of the four response-surface factors, with the offset applied. The interval comes from the t distribution and includes the uncertainty of the coefficients and the residual, the scatter between runs of the scale-down model. It does not include variation between manufacturing batches beyond that scatter, and it does not include the uncertainty of the offset, which the qualification handles (Section 3.1). It is two-sided, so each of its ends is a one-sided 97.5 % limit. A two-sided criterion is judged at both ends. In this report the verdict over the NOR of an attribute means whether it meets its criterion at all 81 combinations.

The PARs were computed as PCMP-001 §6.4 defines them. The set-point analysis holds the other factors at their set-points, scans the factor across its range studied

at 201 points, and takes the contiguous range around the set-point over which the mean prediction meets the criterion. The NOR-propagated analysis scans the factor at 81 points. At each point it draws 2,000 settings of the other factors, each from a normal distribution centred on its set-point with a standard deviation equal to the width of its NOR divided by 6, clipped to the range studied. PCMP-001 §6.4 takes that standard deviation as the routine variation of the parameter. Each simulated run takes its own coefficients from their estimated distribution and its own residual. The 2.5th percentile of the simulated runs must not fall below a lower limit, and the 97.5th percentile must not exceed an upper limit. The NOR-propagated PAR is the one that supports a claim.

The design space was evaluated on a grid of 11 evenly spaced points along each response-surface factor, 14,641 points in all (Section 6). Capability at commercial scale was estimated from the integrated model of the train, which runs 2,000 simulated batches through every step with each parameter drawn as in the NOR-propagated analysis (PCMP-001 §6.5). Cpk is computed against the drug substance criterion, as the lower of the two one-sided indices for a two-sided criterion. The same simulation gives the probability that the drug substance falls outside its criterion, as the count of simulated batches outside it and as the tail probability of a normal distribution with the simulated mean and standard deviation.

PCP-003 §6.4 gives no test for a univariate difference. Each univariate parameter was judged on the difference between the mean of its two runs at the high end and the mean of its two runs at the low end of the range studied, by a t-test against the pooled standard deviation of the duplicate runs (PD-003-05, 8 degrees of freedom), at the level of $\alpha = 0.05$ that PCMP-001 §6.2 sets for a single test. The results of the osmolality study were fitted by a straight line in osmolality, coded -1, 0 and $+1$ at the NOR low, the set-point and the NOR high. Each level was judged on the fitted mean and on the 95 % prediction interval of a single run at manufacturing scale.

4 Study design

4.1 Factors, ranges and the knowledge space

Table 4.1 lists the nine parameters of the step with the set-point, the NOR, the range studied, the study type and the class assigned in Section 9. The set-point is the target of commercial operation, and the NOR is the band within which routine operation holds the parameter. The range studied is the knowledge space of this characterization. No NOR and no set-point of this step was changed at Version 2.0.

Table 4.1: Process parameters of the production bioreactor, with set-point, NOR, range studied (Char. range), class and study type.

Parameter	Unit	Set-point	NOR	Char. range	Class	Study
Culture pH	pH	6.85	6.75–6.95	6.6–7.1	WC-CPP	multivariate
Culture temperature	°C	35	34.5–35.5	34–36	WC-CPP	multivariate
CO2 partial pressure (pCO2)	mmHg	100	70–130	40–160	WC-CPP	multivariate
Osmolality	mOsm/kg	400	375–425	360–440	CPP	multivariate
Culture duration	days	17	16–18	15–19	WC-CPP	multivariate
Dissolved oxygen	% air saturation	50	40–60	30–70	GPP	univariate
Initial viable cell conc.	10^6 cells/mL	1	0.8–1.2	0.5–1.5	KPP	univariate
Basal medium concentration	× standard	1.2	1–1.4	0.8–1.6	KPP	univariate
Nutrient feed-1 volume	% of working volume	12	10–14	9–15	KPP	univariate

Table 4.2 gives the code letter of each factor of the designs. The codes were assigned in screening order and kept for the response-surface terms, so a factor carries the same letter throughout the report.

Table 4.2: Factors of the designs and their code letters.

Code	Factor	Unit	Studied in
A	Culture pH	pH	screening + RSM
B	Culture temperature	°C	screening + RSM
C	CO2 partial pressure (pCO2)	mmHg	screening + RSM
D	Osmolality	mOsm/kg	screening
E	Culture duration	days	screening + RSM

4.2 Screening design

The screening study S-BRX-SCR varied the five multivariate parameters in a resolution V half fraction of the two-level factorial design, with 16 factorial runs and three centre points, 19 runs in all. At resolution V every main effect and every two-factor interaction can be estimated clear of the others, and each is aliased only with interactions of three factors or more. The runs were made in randomized order under SOP-1002. The design carries no squared term, so it can detect curvature but cannot attribute it to a factor. Appendix A gives the settings and the results.

4.3 Response-surface design

The response-surface study S-BRX-RSM varied culture pH, culture temperature, culture duration and CO2 partial pressure (pCO2) in a face-centred central composite design. The design has 16 factorial runs, 8 axial runs at the centres of the faces of the design cube and four centre points, 28 runs in all. Each factor is run at three levels. Osmolality was held at its set-point in every run. PCP-003 §6.3 planned this subset of factors. It also requires the factors carried into the design to be those that the screening identifies as active, with the design amended before execution if the subset differs. Osmolality was active in the screening (Section 5.2) and was not carried forward (PD-003-07). Appendix B gives the settings and the results.

The design was augmented with 16 runs under INV-002, in a D-optimal augmentation run as a second block, the study S-BRX-AUG (2026-09-03 to 2026-09-24) (Fedorov 1972). INV-002 decided its rule on 2026-09-02, after the results of the planned runs were known and before any augmentation run was made. A failing model of an attribute of the drug substance specification on which the planned design shows an effect is augmented, and every other failing model is refitted. The number of runs is the smallest augmentation found in INV-002 to give a probability of at least 0.90 that the full quadratic models of high mannose and acidic variants meet the lack-of-fit test and the limit on overfitting. The augmentation is the discrepancy PD-003-02, and Appendix B gives its runs.

4.4 Univariate assessment

The univariate study S-BRX-UNI (2025-10-06 to 2025-10-31) assessed dissolved oxygen, initial viable cell concentration, basal medium concentration and nutrient feed-1 volume one at a time. Each parameter was run at the low and the high end of its range studied with every other parameter at its set-point, and the harvest was assayed for the five responses and the two impurities. PCP-003 §6.4 plans one run at each end. Duplicate runs were made at each end, 16 runs in all, to measure the variation between runs within the study (PD-003-05). The set-point reference of the study is the scale-down runs of the qualification study. This design cannot show whether the effect of one of these parameters depends on the level of another (International Council for Harmonisation 2009), so the range it supports is conditioned on the other parameters being at their set-points.

4.5 Osmolality study

The osmolality study S-BRX-OSM (2026-09-01 to 2026-09-23) ran osmolality at its NOR low, set-point and NOR high. The four response-surface parameters were held at the combination of NOR settings least favourable for acidic variants, culture pH at pH 6.95, culture temperature at 34.5 °C, culture duration at 18 days and pCO2 at 130 mmHg. The study was made during the Version 2.0 re-analysis, after the screening and response-surface results were known, and it is recorded as the discrepancy PD-003-01. Each level had four runs, the smallest number for which the straight-line prediction at the NOR high has a standard error of at most 0.5 times the standard deviation between runs.

4.6 Execution of the studies

Table 4.3 gives the execution window of each study. Every study of PCP-003 ran in the scale-down bioreactor system whose pCO2 probe is EQ-BRX-205, and each window closed before the next calibration of that probe was due on 2026-06-14. The univariate study started on 2025-10-06, before the response-surface study started on 2025-10-20, against the sequence of PCP-003 §11 (PD-003-10). The records of the osmolality and augmentation studies name no equipment and state no calibration status of the pCO2 probe (PD-003-12). Host cell protein and DNA were measured in the clarified harvest of every run of the screening, response-surface and univariate designs. They were not measured on the qualification runs, on the augmentation runs or in the osmolality study.

Table 4.3: Execution window of each study of the production bioreactor, with the pCO2 probe it used and the next calibration due date of that probe.

Study	Study type	Start	End	pCO2 probe	Calibration due
S-BRX-SDM	Scale-down model qualification	2025-08-11	2025-09-05	EQ-BRX-205	2026-06-14
S-BRX-SCR	Screening design	2025-09-08	2025-10-03	EQ-BRX-205	2026-06-14
S-BRX-UNI	Univariate assessment	2025-10-06	2025-10-31	EQ-BRX-205	2026-06-14
S-BRX-RSM	Response-surface design	2025-10-20	2025-12-12	EQ-BRX-205	2026-06-14
S-BRX-OSM	Osmolality study	2026-09-01	2026-09-23	not recorded (PD-003-12)	not recorded (PD-003-12)
S-BRX-AUG	Augmentation (INV-002)	2026-09-03	2026-09-24	not recorded (PD-003-12)	not recorded (PD-003-12)

5 Results

5.1 Centre-point performance and reproducibility

Table 5.1 gives the mean, the standard deviation and the coefficient of variation of the centre points of both designs. These replicates carry the variation between runs of the scale-down model and of the methods, and they are the pure error of the lack-of-fit test. High mannose is the least precise attribute at the centre point in both designs, with a coefficient of variation of 11.5 % in the screening design and 9.9 % in the response-surface design. That is several times the precision of the glycan map, 2.5 % of the result (Appendix D), so most of the variation in high mannose comes from the culture. The standard deviation of galactosylation at the centre point is 2.34 points in the response-surface design against 0.48 in the screening design. The larger value enters the lack-of-fit test of the galactosylation model as pure error, which is one reason why that test has the highest p-value of the five models (Section 5.3.1).

Table 5.1: Mean, standard deviation and coefficient of variation of the centre points of the two designs, as measured in the scale-down model.

Attribute	Screening design				Response-surface design			
	n	Mean	SD	CV (%)	n	Mean	SD	CV (%)
Afucosylation (%)	3	7.93	0.50	6.3	4	7.61	0.35	4.7
Galactosylation (%)	3	26.9	0.48	1.8	4	25.9	2.34	9.0
High mannose (%)	3	6.28	0.72	11.5	4	5.66	0.56	9.9
Acidic variants (%)	3	21.3	1.09	5.1	4	21.5	0.88	4.1
Aggregates, HMW (%)	3	1.59	0.08	4.7	4	1.51	0.10	6.5

5.2 Screening: factor effects

Table 5.2 lists the effects significant at $\alpha = 0.05$ for each attribute, as the change in the attribute across the range studied of the factor. The screening models of afucosylation and high mannose have no significant model F-test ($p = 0.092$ and $p = 0.183$), so their individual effects identify candidates and no more. No attribute shows significant curvature in the screening design, with lack-of-fit p-values from 0.078 to 0.688, but with two degrees of freedom for pure error that test detects only a large curvature.

pCO2 has the largest effect on acidic variants, which fall by 8.21 points from the low to the high end of its range. A higher culture pH also lowers acidic variants, and a higher culture temperature raises them. Culture duration has the largest effect on

galactosylation, a fall of 7.86 points, and the interaction of culture pH with culture duration is nearly as large and of opposite sign. A higher pCO2 and a higher culture pH lower galactosylation, and a higher culture temperature raises it. A higher culture pH and a longer culture raise aggregate, by 0.83 and 0.58 points. A higher culture pH raises high mannose and a higher culture temperature lowers it. Afucosylation falls with a longer culture, and its interactions of culture pH with culture duration and with pCO2 are significant.

A higher osmolality lowers galactosylation by 2.50 points (p = 0.008) and acidic variants by 2.61 points (p = 0.012) across its range, and its interactions with culture temperature, pCO2 and culture duration are significant for galactosylation. Under PCP-003 §6.3 osmolality should therefore have been carried into the response-surface design, with the design amended before execution. It was not (PD-003-07). The osmolality study later measured its effect at the least favourable NOR combination for acidic variants (Section 5.5).

Screening run 4 ran below its design setting of pCO2, because the pCO2 probe read high (study event DEV-003-01). Appendix A shows the run at its design setting, and the analysis uses its recorded setting of 153.5 mmHg. The set of significant effects in Table 5.2 is the same with the run at its design setting (Section 13.1). The response-surface model re-estimates the effects of the four factors carried forward, in a design without aliases, and it is the model from which the ranges and the design space are predicted.

Table 5.2: Effects significant at α = 0.05 in the screening design. The effect is the change in the attribute across the range studied of the factor, as measured in the scale-down model.

Attribute	Term	Effect (points)	p-value
Afucosylation (%)	Culture pH × culture duration	2.45	0.019
	Culture duration	−2.37	0.021
	Culture pH × pCO2	−1.77	0.045
Galactosylation (%)	Culture duration	−7.86	< 0.001
	Culture pH × culture duration	6.95	< 0.001
	pCO2	−5.84	< 0.001
	Culture pH	−4.51	0.001
	Culture temperature × osmolality	3.75	0.003
	Culture pH × pCO2	3.65	0.003
	Culture temperature	3.59	0.003
	Culture pH × culture temperature	3.23	0.004
	Osmolality	−2.50	0.008
	pCO2 × osmolality	2.13	0.013
	Osmolality × culture duration	1.88	0.018
	Culture temperature × culture duration	1.54	0.030
High mannose (%)	Culture pH	1.52	0.016
	Culture temperature	−1.39	0.021
Acidic variants (%)	pCO2	−8.21	< 0.001
	Culture pH	−4.11	0.003
	Culture temperature	3.02	0.008
	Osmolality	−2.61	0.012
	Culture pH × culture temperature	1.93	0.027
	Culture pH × pCO2	−1.87	0.029
	Culture temperature × culture duration	1.78	0.033
Aggregates, HMW (%)	Culture pH	0.83	< 0.001
	Culture duration	0.58	< 0.001
	Culture temperature × culture duration	−0.14	0.042

5.3 Response-surface models

5.3.1 Selected models and their adequacy

Table 5.3 gives the selected model of each attribute on all 44 runs. The raw scale is used for afucosylation and galactosylation, and the log scale for high mannose, acidic variants and aggregate. For afucosylation the Box-Cox interval excludes both 0 and

1, so the rule keeps the raw scale although the profile does not support it either (Section 5.3.2). Every selected model meets the model F-test ($p < 0.001$ for every model), the lack-of-fit test (p from 0.246 to 0.994) and the limit on overfitting, with predicted R^2 at most 0.050 below R^2 against the limit of 0.20.

Table 5.3: Selected response-surface models of the production bioreactor on all 44 runs, against the model F-test, the lack-of-fit test and the limit on overfitting.

Attribute	Scale	Box-Cox λ (95 % interval)	Terms	R^2	Adj. R^2	Pred. R^2	R^2 minus pred. R^2	F-test p	Lack of fit p
Afucosylation (%)	raw	0.44 (0.11 to 0.79)	9	0.955	0.941	0.917	0.038	< 0.001	0.246
Galactosylation (%)	raw	0.77 (0.13 to 1.37)	7	0.960	0.951	0.939	0.022	< 0.001	0.994
High mannose (%)	log	−0.86 (−2.00 to 0.30)	3	0.824	0.806	0.774	0.050	< 0.001	0.820
Acidic variants (%)	log	−0.44 (−1.04 to 0.22)	9	0.944	0.928	0.897	0.047	< 0.001	0.295
Aggregates, HMW (%)	log	0.13 (−0.43 to 0.76)	4	0.955	0.949	0.939	0.016	< 0.001	0.514

Terms are counted without the intercept and the block term. Adj. R^2, adjusted R^2. Pred. R^2, predicted R^2 from PRESS. The model scale is the log scale where the 95 % interval of the Box-Cox exponent λ contains 0 and excludes 1, and the raw scale otherwise.

At the set-points, the selected models of afucosylation and acidic variants predict 8.29 % and 22.23 % in the scale-down model. The means of the four centre points of the response-surface design are 7.61 % and 21.50 %, with standard errors of 0.18 and 0.44 points. The lack-of-fit test is not significant for either model. At manufacturing scale the model of acidic variants predicts 22.83 % at the set-point, against a geometric mean of 21.85 % in the manufacturing batches of the qualification. For acidic variants the difference lies in the unfavourable direction, because a model that predicts too high near the set-point overstates the distance to the lower limit. The predicted median at the least favourable NOR combination lies 0.39 points above the lower limit, less than either difference at the set-point. If the model overstates acidic variants there by the same factor as at the centre points, the median at that combination is about 17.79 %, below the lower limit (Section 7.2).

Table 5.4 sets the plan's full quadratic model, fitted on the raw scale to the planned runs and to all runs with a block term, beside the model used. On the planned runs the full quadratic models of high mannose and acidic variants exceed the limit on overfitting, which is why the design was augmented. On all runs the full quadratic model of acidic variants meets the limit, but that of high mannose still exceeds it, with predicted R^2 0.25 below R^2. The selected model of high mannose meets the limit, and the full quadratic model that the plan prescribes would leave high mannose without an adequate model on all runs.

Table 5.4: Re-evaluation of the response-surface models with the augmentation runs.

Attribute	Planned runs, full quadratic				All runs, full quadratic with block						Model used (terms)
	Runs	R^2	Pred. R^2	Gap	Runs	R^2	Pred. R^2	Gap	Lack of fit p	Block p	
Afucosylation (%)	28	0.96	0.79	0.17	44	0.96	0.90	0.06	0.237	0.377	9 terms
Galactosylation (%)	28	0.97	0.92	0.06	44	0.96	0.92	0.04	0.990	0.518	7 terms
High mannose (%)	28	0.91	0.65	0.26	44	0.84	0.59	0.25	0.638	0.898	Culture pH, culture temperature, culture pH × culture temperature, log scale
Acidic variants (%)	28	0.94	0.60	0.34	44	0.93	0.81	0.12	0.188	0.640	9 terms, log scale
Aggregates, HMW (%)	28	0.97	0.83	0.13	44	0.95	0.87	0.08	0.415	0.096	4 terms, log scale

Gap: R^2 minus predicted R^2, at most 0.20 under PCMP-001 §7.2. Model used: the scale chosen by the Box-Cox profile and the model with the lowest AICc among the hierarchical submodels of the full quadratic, a block term kept (INV-002).

5.3.2 Residual patterns and influential runs

The selected models of afucosylation, high mannose and aggregate meet the criterion on residuals and influential runs, and those of galactosylation and acidic variants do not (Table 5.5). For these two the absolute residuals rise with the predicted value, with a slope significant at $p = 0.027$ for galactosylation and $p < 0.001$ for acidic variants. No model shows a trend with the run order, and the lowest p-value of that test is 0.720. No run has a Cook's distance above the F median of its model, which lies between 0.886 and 0.959 for these models. Every model has at least one run with a Cook's distance above $4/n = 0.091$. Across the five models there are 17 such cases, on 13 different runs.

Each of these runs was deleted in turn and the model selection repeated. For every attribute and every deleted run, the model selected without the run met the model F-test, the lack-of-fit test and the limit on overfitting, every attribute kept its verdict over the NOR, and every NOR-propagated PAR still contained the NOR. The share of the region studied that lies inside the design space ranged from 73.5 % to 75.5 %, against 73.7 % with all runs. The same model was selected in 12 of the 17 cases. For afucosylation, without run 1 the selection adds culture duration × pCO2 and $pCO2^2$. For high mannose, without run 34 the raw scale is selected with the same terms. For acidic variants, without run 1 the selection drops culture temperature × pCO2, without run 3 the selection drops culture pH × pCO2 and without run 30 the selection drops culture temperature × pCO2.

Table 5.5: Residual patterns and influential runs of the selected models, against the criterion on residuals and influential runs.

Attribute	Spread p	Order p	Largest Cook's distance (run)	F median	Runs above 4/n	Verdict
Afucosylation (%)	0.062	0.720	0.289 (1)	0.959	1, 5, 36, 43	met
Galactosylation (%)	0.027	0.776	0.346 (33)	0.945	5, 10, 33	not met
High mannose (%)	0.164	0.857	0.243 (10)	0.886	10, 11, 15, 34	met
Acidic variants (%)	< 0.001	0.882	0.303 (3)	0.959	1, 3, 30, 36	not met
Aggregates, HMW (%)	0.291	0.926	0.141 (29)	0.907	29, 32	met

Spread, the regression of the absolute residuals on the predicted values. Order, the regression of the residuals on the run order, the augmentation runs continuing the planned order. A run above $4/n = 0.091$ for 44 runs is examined by deletion, and the model fails the criterion only on a significant slope or on a run above the F median, the median of the F distribution with p and $n - p$ degrees of freedom, where p is the number of parameters of the model.

Each residual trend was sized against the criterion of its attribute, on the residual standard deviation in the lower and the upper half of the predicted values. For galactosylation it is 1.07 points in the lower half and 1.09 points in the upper half. The rise that the slope detects therefore does not show as a difference between the halves, and both are small against the width of the criterion, 30 points. For acidic variants it is 0.033 in the lower half and 0.063 in the upper half, on the log scale. With the standard deviation of the half in which each prediction lies, both attributes keep their verdict over the NOR. For acidic variants the lowest prediction limit moves from 16.25

% to 17.02 %, still below the lower limit. Afucosylation, whose Box-Cox profile rejects the raw scale and whose slope is not significant (p = 0.062), has a residual standard deviation of 0.34 points in the lower half and 0.58 in the upper half. Judged with the standard deviation of each half, it still meets its criterion across the NOR, with prediction limits from 6.13 % to 11.39 %.

The normal quantile plots of the residuals in Figure 5.1 to Figure 5.5 show no departure from normality, so the normality criterion is met.

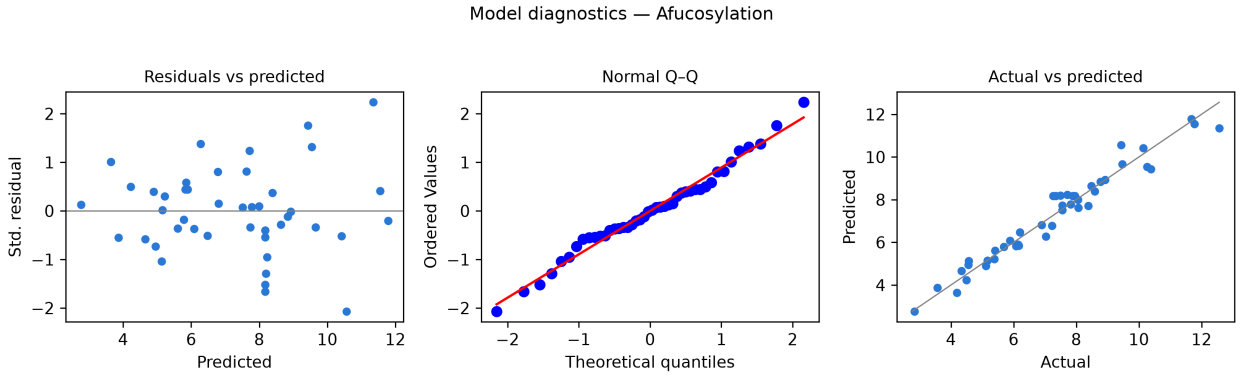


Figure 5.1: Residual diagnostics of the selected model of afucosylation, on the raw scale. Left, standardized residuals against predicted values. Centre, normal quantile plot. Right, predicted against measured values.

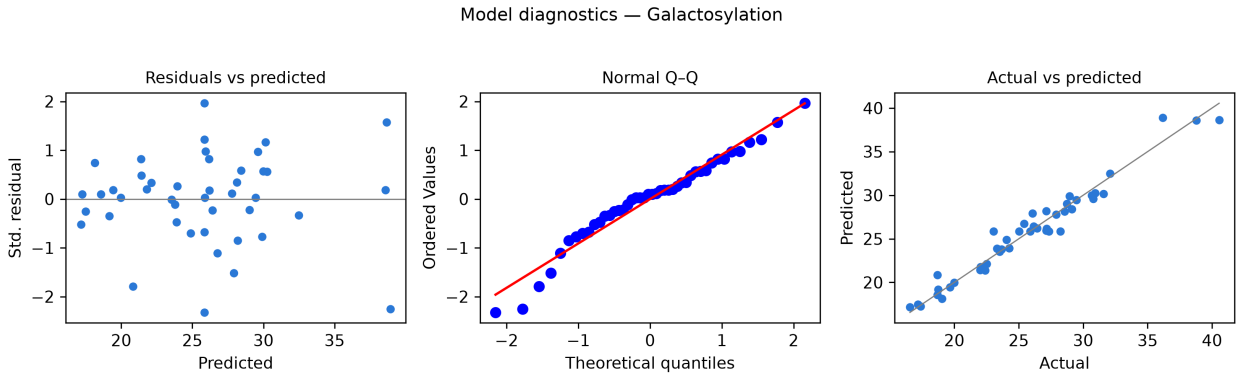


Figure 5.2: Residual diagnostics of the selected model of galactosylation, on the raw scale. Left, standardized residuals against predicted values. Centre, normal quantile plot. Right, predicted against measured values.

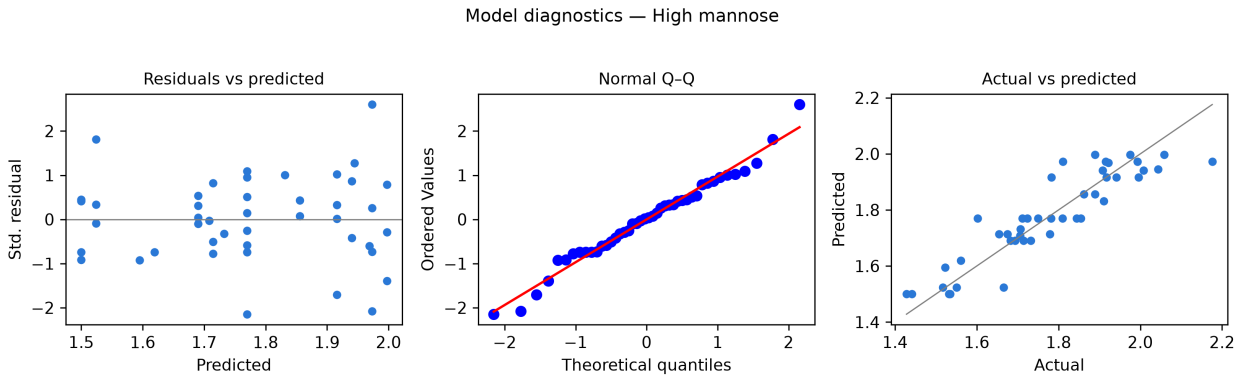


Figure 5.3: Residual diagnostics of the selected model of high mannose, on the log scale. Left, standardized residuals against predicted values. Centre, normal quantile plot. Right, predicted against measured values.

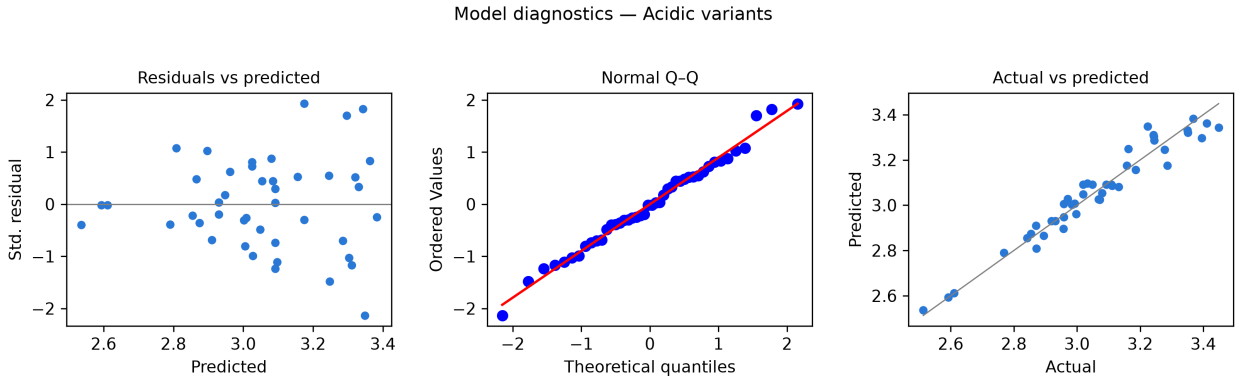


Figure 5.4: Residual diagnostics of the selected model of acidic variants, on the log scale. Left, standardized residuals against predicted values. Centre, normal quantile plot. Right, predicted against measured values.

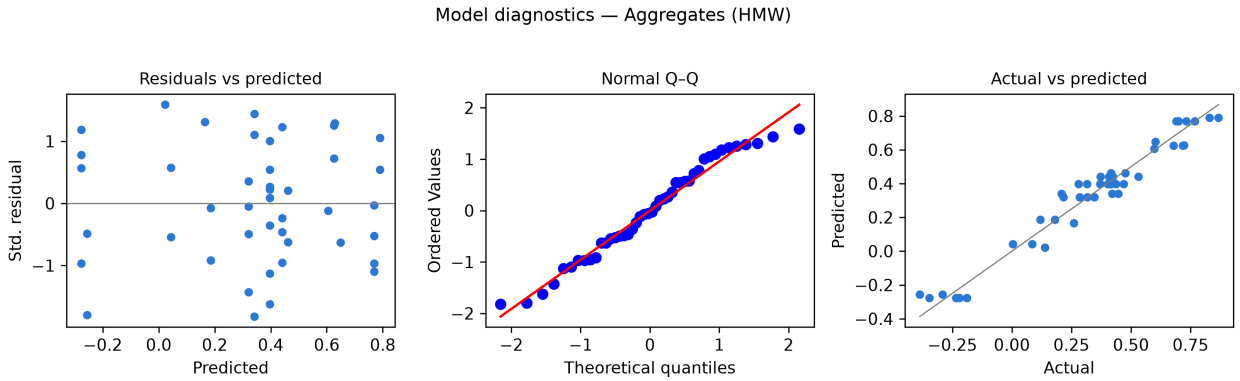


Figure 5.5: Residual diagnostics of the selected model of aggregate, on the log scale. Left, standardized residuals against predicted values. Centre, normal quantile plot. Right, predicted against measured values.

5.3.3 Effects of the parameters

The coefficients of the selected models are in coded units, so each measures the change from the set-point to the end of the range of its factor. Appendix C gives the coefficient tables of afucosylation, galactosylation, high mannose and aggregate, and Table 5.7 and Table 5.8 those of acidic variants, the attribute nearest its limit. Table 5.6 states each main effect as the change it causes across the range studied, at manufacturing scale with the other factors at their set-points, against the width of the criterion.

PCP-003 §7 calls an effect practically important when that change is large relative to the width of the criterion, and PCMP-001 §7.7 asks for the size of an effect beside its significance. Neither sets a threshold. Table 5.6 describes a change of a fifth of the width or more as large, of a tenth to a fifth as moderate and of less than a tenth as small. These bands are a judgement of this report, and no class rests on them (Section 9). Only for acidic variants do the effects bring the attribute to its limit within the NOR. There a higher pCO2 and a higher culture pH each lower acidic variants, and the PARs of these two parameters end just above their NOR highs (Section 7.3).

Table 5.6: Main effects of the selected models against the width of each criterion, at manufacturing scale with the other factors at their set-points.

Attribute	Parameter	Change over the range studied	Change over the NOR	Criterion width	Share of the width (%)	Size
Afucosylation (%)	Culture pH	−1.94	−0.78	11	17.7	moderate
	Culture temperature	−1.32	−0.66	11	12.0	moderate
	Culture duration	−2.73	−1.37	11	24.8	large
	pCO2	−0.92	−0.46	11	8.4	small
Galactosylation (%)	Culture pH	−4.12	−1.65	30	13.7	moderate
	Culture temperature	3.83	1.91	30	12.8	moderate
	Culture duration	−7.48	−3.74	30	24.9	large
	pCO2	−4.63	−2.32	30	15.4	moderate
High mannose (%)	Culture pH	2.25	0.90	7	32.1	large
	Culture temperature	−0.79	−0.40	7	11.3	moderate
Acidic variants (%)	Culture pH	−4.56	−1.88	21	21.7	large
	Culture temperature	3.80	1.90	21	18.1	moderate
	Culture duration	−0.27	−0.13	21	1.3	small
	pCO2	−8.93	−4.44	21	42.5	large
Aggregates, HMW (%)	Culture pH	0.89	0.38	6.3	14.1	moderate
	Culture duration	0.77	0.38	6.3	12.1	moderate

Main effects only, for the parameters in each selected model. Changes and widths in percentage points. Size, the report's own bands: large from a fifth of the width, moderate from a tenth, small below a tenth. The aggregate criterion has an upper limit only, and its width is taken from 0.

Afucosylation falls as culture duration, culture pH, culture temperature and pCO2 rise. The interaction of culture pH with culture duration is as large as the main effect of culture duration and of opposite sign. At the low end of the pH range a longer culture lowers afucosylation by 5.43 points across the range of culture duration, and at the high end of the pH range it changes it by 0.03 points. The squared term of

culture pH is negative, so with the other factors at their set-points afucosylation is highest inside the pH range and falls toward both ends.

Galactosylation falls with culture duration, pCO2 and culture pH, and rises with culture temperature. Culture pH changes the size of each of these effects. Across the range of culture duration galactosylation falls by 12.7 points at the low end of the pH range and by 2.3 points at the high end. Across the range of pCO2 it falls by 8.7 and 0.6 points. Across the range of culture temperature it rises by 0.1 points at the low end of the pH range and by 7.6 points at the high end.

High mannose rises with culture pH, by a factor of 1.42 across its range, and falls with culture temperature, by a factor of 0.88. The small interaction of the two is significant. Neither culture duration nor pCO2 is in the model.

Acidic variants fall with pCO2, by a factor of 0.68 across its range, and with culture pH, by a factor of 0.81. They rise with culture temperature, by a factor of 1.18. The main effect of culture duration is a factor of 0.99 and is not significant ($p = 0.583$). It stays in the model because the interaction of culture pH with culture duration does. The interaction of culture pH with culture temperature is significant. Of the terms of this model, only culture pH, culture temperature, pCO2 and culture pH × culture temperature are carried by every model whose AICc lies within 2 units of the lowest (Section 5.3.5).

Aggregate rises with culture pH, by a factor of 1.79 across its range, and with culture duration, by a factor of 1.59. The interaction of the two is negative on the log scale, so a longer culture multiplies aggregate by less at high pH than at low pH. The squared term of culture pH is negative, so the rise with pH flattens toward the top of the range. Neither culture temperature nor pCO2 is in the model.

Figure 5.6 shows the response surfaces over culture pH and culture duration, with culture temperature and pCO2 at their set-points. The surfaces are predictions of the scale-down model before the offset, and for a log-scale attribute they show the median.

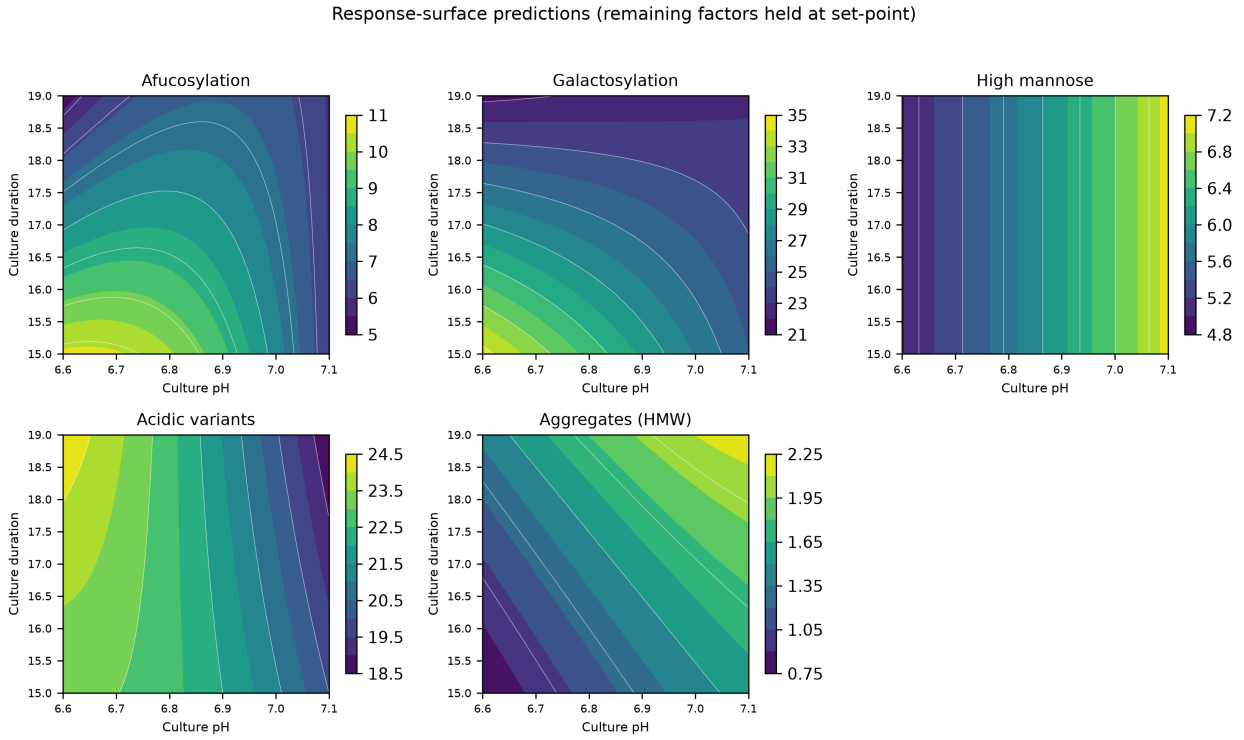


Figure 5.6: Response-surface predictions of the modelled attributes over culture pH and culture duration, with culture temperature and pCO2 at their set-points. Values of the scale-down model before the offset, and the median for an attribute modelled on the log scale.

Table 5.7: Coefficients of the selected model of acidic variants, on the log scale and in coded units. The factor codes are in Table 4.2.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	3.1012	0.0170	182.0418	< 0.001	***
A	−0.1038	0.0105	−9.8722	< 0.001	***
B	0.0831	0.0104	8.0137	< 0.001	***
E	−0.0059	0.0107	−0.5539	0.583	
C	−0.1943	0.0104	−18.7457	< 0.001	***
AB	0.0565	0.0113	5.0156	< 0.001	***
AE	−0.0319	0.0117	−2.7209	0.010	*
AC	−0.0206	0.0113	−1.8290	0.076	
BC	0.0197	0.0110	1.7803	0.084	
A^2	−0.0395	0.0197	−2.0090	0.053	
Block	0.0100	0.0094	1.0664	0.294	

Table 5.8: Analysis of variance of the selected model of acidic variants, on the log scale, with the residual split into lack of fit and pure error.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	1.8999	10	0.1900	56.0751	< 0.001
Residual	0.1118	33	0.0034		
Lack of fit	0.1068	30	0.0036	2.1271	0.295
Pure error	0.0050	3	0.0017		

5.3.4 What the augmentation changed

The 16 augmentation runs were added after the results of the planned runs were known, and every selected model is fitted to them. Their pCO2 settings rest on a probe whose calibration status the record does not give (PD-003-12).

On the 28 planned runs alone, the selected models meet the model F-test, the lack-of-fit test and the limit on overfitting for every attribute. Their NOR-propagated PARs for acidic variants end at pH 6.93 for culture pH, start at 34.6 °C for culture temperature and end at 116 mmHg for pCO2, so none of the three contains the NOR. At the least favourable NOR combination the predicted median is 18.01 % and the lower prediction limit 15.48 %, and the design space covers 59.6 % of the region studied. With the plan's full quadratic model on all 44 runs, the PAR of pCO2 ends at 122 mmHg, the design space covers 63.6 %, and high mannose exceeds the limit on overfitting. That every PAR contains the NOR, and that the design space covers 73.7 %, therefore rests on the augmentation runs and on the model selection together.

A model selected on the same runs on which it is judged passes an optimistic check, and the augmentation runs are an independent one. Table 5.9 shows how well the models selected on the planned runs predict the augmentation runs. The models of high mannose and aggregate predict them poorly, with a root mean square error of prediction 1.7 and 1.8 times their residual standard deviation. For aggregate only 11 of the 16 augmentation runs fall inside their 95 % prediction intervals, so the single-run intervals of aggregate understate the scatter between runs. That limits the verdict of aggregate over the NOR. The block term is not significant for any attribute ($p \geq 0.20$), so a shift between the two blocks does not explain the misses.

Table 5.9: Prediction of the augmentation runs by the models selected on the planned runs alone.

Attribute	Scale, planned runs to all runs	Augmentation runs inside the 95 % prediction interval	RMSEP over residual SD	Block p
Afucosylation (%)	log to raw	14 of 16	1.30	0.201
Galactosylation (%)	raw to raw	15 of 16	1.11	0.497
High mannose (%)	raw to log	13 of 16	1.70	0.329
Acidic variants (%)	log to log	16 of 16	0.92	0.294
Aggregates, HMW (%)	log to log	11 of 16	1.81	0.382

RMSEP, the root mean square error of the predictions of the augmentation runs, over the residual standard deviation of the same model on the planned runs, both on the scale of that model. Block p, the block term of the model selected on all runs.

On all runs, the model of high mannose moves from the raw to the log scale, loses culture duration, pCO2, culture pH × culture duration and culture pH × pCO2, and gains culture pH × culture temperature. The model of aggregate loses culture temperature and culture temperature², and gains culture pH². The model of acidic variants gains culture duration, culture pH × culture duration, culture temperature × pCO2 and culture pH², and that of afucosylation moves from the log to the raw scale. Only the model of galactosylation is the same. The augmentation therefore changed what four of the five models say, and for high mannose the interactions that the planned runs suggested are not supported by all runs.

5.3.5 Near-equivalent models

Besides the selected models, 15 models have an AICc within 2 units of the lowest, 2 for afucosylation, none for galactosylation, 2 for high mannose, 10 for acidic variants and 1 for aggregate. Each of them meets the model F-test, the lack-of-fit test and the limit on overfitting, gives every attribute the same verdict over the NOR as the selected model and leaves the scale-down model qualified. They differ from the selected model by one to four terms. For afucosylation some carry culture duration × pCO2 and pCO2², for high mannose pCO2 and culture pH × pCO2, and for aggregate pCO2 and pCO2². For acidic variants they share only culture pH, culture temperature, pCO2 and culture pH × culture temperature.

5.4 Univariate assessment

Table 5.10 gives every result of the univariate study for the five attributes, and Table 5.11 the difference between the two ends of each range, both at manufacturing scale. Every result meets its criterion. The half-width of the 95 % confidence interval in Table 5.11 is the precision of each comparison. For galactosylation, for example, a difference of 3.1 points between the two ends of a range is detected about half the time, and a difference of 4.3 points with a probability of 80 %.

Acidic variants are 3.51 points higher at the high end of the initial viable cell concentration than at the low end, with a 95 % confidence interval of ± 2.36 points (p = 0.009). It is the only significant result among the 28 univariate tests of Table 5.11. Each was made at the level that PCMP-001 §6.2 sets for a single test, at which about one test in twenty with no real effect gives a significant result, so about 1.4 such results are expected among them by chance. No correction for the number of tests is applied, because none was set before the result. No effect of dissolved oxygen, basal medium concentration or nutrient feed-1 volume was detected on any attribute, at the precision given in Table 5.11, and no effect of the initial viable cell concentration was detected on any other attribute.

PCP-003 §6.4 carries a univariate parameter into a multivariate study when it shows an effect on a quality attribute that the mechanism given in PCP-003 §4.3 does not account for. That mechanism, restated in Section 2.1, expects the initial viable cell concentration to change the number of viable cells and the titre, and no quality attribute. One route to acidic variants would be the time the antibody spends in the culture fluid at culture temperature, since a culture that starts at a higher cell concentration makes its antibody earlier. That route does not account for the effect. Culture duration sets the time in the culture fluid most directly, and it has no detectable main effect on acidic variants, in the screening design (p = 0.956) or in the response-surface model (p = 0.583). The parameter should therefore have been carried into a multivariate study and the plan amended, and that was not done (PD-003-11).

A straight line between the two ends of the range puts acidic variants 0.70 points lower at the NOR low of the initial viable cell concentration than at its set-point, with the other parameters at their set-points. The response-surface and osmolality studies ran at the set-point of this parameter, so the verdict over the NOR in Section 7.2 does not include its effect. Added to the least favourable combination of the four response-surface parameters, on the assumption that the effect does not depend on them, it would move the predicted median from 18.39 % to 17.69 %, below the lower limit, and the lower prediction limit from 16.25 % to 15.55 %. The acceptance of the acidic-variant risk in Section 7.2 covers this combination on the same grounds. Acidic variants at harvest are to be trended against the initial viable cell concentration of each production batch, so that a false positive is found out.

Table 5.10: Results of the univariate study S-BRX-UNI for the five modelled attributes, at manufacturing scale.

Parameter	End of range	Run	Afuc. (%)	Gal. (%)	High man. (%)	Acidic (%)	HMW (%)
Dissolved oxygen (% air saturation)	Low, 30	1	8.36	26.9	6.01	21.3	1.52
		2	7.95	28.6	6.54	22.9	1.59
	High, 70	1	8.64	26.3	6.62	22.1	1.72
		2	8.73	28.6	6.95	23.7	1.56
Initial viable cell conc. (10^6 cells/mL)	Low, 0.5	1	9.03	26.8	6.72	20.1	1.51
		2	8.48	24.7	6.57	21.3	1.89
	High, 1.5	1	8.61	27.6	6.99	24.2	1.72
		2	8.52	25.6	7.44	24.3	1.49
Basal medium concentration (× standard)	Low, 0.8	1	8.32	27.2	6.56	24.0	1.83
		2	7.54	27.2	6.72	22.4	1.70
	High, 1.6	1	8.92	23.1	6.69	21.1	1.51
		2	8.34	25.8	6.15	22.8	1.70
Nutrient feed-1 volume (% of working volume)	Low, 9	1	8.40	25.0	6.38	21.0	1.61
		2	7.56	26.7	6.69	23.2	1.59
	High, 15	1	7.43	27.1	6.43	22.9	1.71
		2	8.24	25.6	5.96	22.9	1.82

Afuc., afucosylation. Gal., galactosylation. High man., high mannose. Acidic, acidic variants. HMW, aggregate. The criteria of the step are afucosylation 2–13 %, galactosylation 10–40 %, high mannose 3–10 %, acidic variants 18–39 % and aggregate ≤ 6.3 %.

Table 5.11: Difference between the high and the low end of each univariate parameter. The five modelled attributes at manufacturing scale, host cell protein and DNA in the clarified harvest as measured.

Parameter	Attribute	Mean, low end	Mean, high end	High minus low	95 % CI half-width	p-value
Dissolved oxygen	Afucosylation (%)	8.16	8.68	0.53	0.97	0.245
	Galactosylation (%)	27.8	27.4	−0.32	3.10	0.819
	High mannose (%)	6.27	6.78	0.51	0.65	0.106
	Acidic variants (%)	22.1	22.9	0.82	2.36	0.449
	Aggregates, HMW (%)	1.56	1.64	0.08	0.31	0.559
	Host cell protein (ng/mg)	615,000	549,000	−65,800	243,000	0.550
	DNA (pg/mg)	187,000	223,000	36,000	145,000	0.582
Initial viable cell conc.	Afucosylation (%)	8.76	8.56	−0.19	0.97	0.657
	Galactosylation (%)	25.8	26.6	0.84	3.10	0.548
	High mannose (%)	6.64	7.22	0.57	0.65	0.075
	Acidic variants (%)	20.7	24.2	3.51	2.36	0.009
	Aggregates, HMW (%)	1.70	1.61	−0.09	0.31	0.518
	Host cell protein (ng/mg)	446,000	621,000	175,000	243,000	0.136
	DNA (pg/mg)	273,000	239,000	−34,500	145,000	0.598
Basal medium concentration	Afucosylation (%)	7.93	8.63	0.70	0.97	0.135
	Galactosylation (%)	27.2	24.4	−2.75	3.10	0.075
	High mannose (%)	6.64	6.42	−0.22	0.65	0.454
	Acidic variants (%)	23.2	21.9	−1.30	2.36	0.240
	Aggregates, HMW (%)	1.77	1.61	−0.16	0.31	0.266
	Host cell protein (ng/mg)	602,000	584,000	−18,500	243,000	0.865
	DNA (pg/mg)	251,000	240,000	−11,600	145,000	0.858
Nutrient feed-1 volume	Afucosylation (%)	7.98	7.83	−0.14	0.97	0.740
	Galactosylation (%)	25.9	26.4	0.50	3.10	0.718
	High mannose (%)	6.54	6.19	−0.35	0.65	0.252
	Acidic variants (%)	22.1	22.9	0.75	2.36	0.482
	Aggregates, HMW (%)	1.60	1.76	0.16	0.31	0.262
	Host cell protein (ng/mg)	740,000	534,000	−206,000	243,000	0.087
	DNA (pg/mg)	218,000	233,000	15,600	145,000	0.810

Each mean is of two runs. The confidence interval uses the pooled standard deviation of the duplicate runs of the study, with 8 degrees of freedom. The p-value is the same at either scale.

5.5 Osmolality study

Table 5.12 gives each attribute at the three levels of osmolality, at manufacturing scale, with the other parameters at the combination of NOR settings least favourable for acidic variants. The means in the table are values of the straight line fitted to the 12 runs. For acidic variants the line falls by 1.48 points across the NOR of osmolality, but its slope is not significant (p = 0.191). Its fitted mean lies above the lower limit at the NOR low and at the set-point, and below it at the NOR high, at 17.7 %. The prediction interval of a single run reaches below the lower limit at every level, as

the response-surface model showed at the set-point of osmolality. Afucosylation falls by 0.95 points across the NOR, with a significant slope (p = 0.023), and galactosylation by 2.21 points (p = 0.051). Every attribute other than acidic variants meets its criterion at every level.

The study therefore shows no significant effect of osmolality on acidic variants at this combination, and the fitted mean at the NOR high is not a finding. It ran at that single combination, with osmolality set by sodium chloride independently of pCO2, so it does not give the interactions of osmolality with the other parameters. Its record names no equipment, so the calibration of the probe that held pCO2 at 130 mmHg is not documented (PD-003-12). The study was decided after the screening and response-surface results were known.

Table 5.12: Results of the osmolality study S-BRX-OSM at manufacturing scale, with the other parameters at the combination of NOR settings least favourable for acidic variants.

Attribute	Criterion (%)	Change across the NOR (p)	Osmolality (mOsm/kg)	Fitted mean	95 % prediction interval	Fitted mean meets	Interval meets
Afucosylation (%)	2–13	−0.95 (0.023)	375	7.73	6.50 to 8.96	Yes	Yes
			400	7.26	6.09 to 8.42	Yes	Yes
			425	6.78	5.55 to 8.01	Yes	Yes
Galactosylation (%)	10–40	−2.21 (0.051)	375	23.0	19.5 to 26.4	Yes	Yes
			400	21.9	18.6 to 25.1	Yes	Yes
			425	20.8	17.3 to 24.2	Yes	Yes
High mannose (%)	3–10	0.20 (0.710)	375	7.03	5.25 to 8.80	Yes	Yes
			400	7.12	5.44 to 8.81	Yes	Yes
			425	7.22	5.44 to 9.00	Yes	Yes
Acidic variants (%)	18–39	−1.48 (0.191)	375	19.1	15.5 to 22.8	Yes	No
			400	18.4	15.0 to 21.9	Yes	No
			425	17.7	14.0 to 21.3	No	No
Aggregates, HMW (%)	≤ 6.3	−0.02 (0.852)	375	1.97	1.68 to 2.26	Yes	Yes
			400	1.96	1.69 to 2.23	Yes	Yes
			425	1.95	1.66 to 2.24	Yes	Yes

All values at manufacturing scale and in percentage points of the attribute. Change across the NOR, from the NOR low to the NOR high of osmolality, from a straight line through the 12 runs, with the p-value of its slope. Both ends of a two-sided criterion are judged.

5.6 Mechanistic interpretation

Galactosylation falls with culture duration, the direction that the late depletion of UDP-galactose and of the manganese cofactor of the galactosyltransferase predicts (Section 2.1). The fall is much larger at the low end of the pH range than at the high end. A galactosyltransferase whose rate depends on the pH of the Golgi predicts an interaction of culture pH with culture duration, but not its sign. Culture temperature raises galactosylation at high pH and has no effect at low pH.

Afucosylation also falls with culture duration at low pH, and hardly at high pH. The mechanism predicts a rise. A pool weighted toward antibody made late in the culture,

when GDP-fucose and the processing capacity of the Golgi have declined, should carry more antibody without core fucose.

A higher pCO2 lowers galactosylation, afucosylation and acidic variants, in each case the direction of a higher culture pH. The mechanism predicts the direction of a lower culture pH, because dissolved carbon dioxide lowers the pH in the cytosol. Osmolality was a factor of the screening design, at 360 and 440 mOsm/kg, set independently of pCO2, and it was held at its set-point in the response-surface design. A rise of osmolality with pCO2 therefore does not explain the direction.

Acidic variants rise with culture temperature, the direction in which a higher temperature speeds deamidation. They fall as culture pH rises, although a higher pH speeds deamidation too. A higher pH would have to lower the sialylated or glycated forms by more than it raises the deamidated ones, and icIEF does not measure the three routes apart.

Aggregate rises with culture pH and with culture duration, the directions that the weaker repulsion between antibody molecules near the isoelectric point and the longer time at temperature predict (Section 2.1). On the log scale the two effects multiply, so aggregate is highest after a long culture at high pH. High mannose rises with culture pH and falls with culture temperature. A warmer culture speeds the processing enzymes of the Golgi, which predicts less high mannose, and the mechanism gives no direction for the pH effect.

The mechanism of Section 2.1 therefore does not explain the direction of the effect of culture duration on afucosylation, of pCO2 on galactosylation, afucosylation and acidic variants, or of culture pH on acidic variants, and it does not account for the effect of the initial viable cell concentration on acidic variants (Section 5.4).

For the operation of the step, the attribute nearest its limit is acidic variants at its lower limit. A high culture pH, a high pCO2 and a low culture temperature lower acidic variants. A high osmolality lowered them in the screening design, and a low initial viable cell concentration in the univariate study. Galactosylation approaches its upper limit at low pH, high temperature, short culture and low pCO2, but at the least favourable NOR combination its prediction interval stays 5.9 points below that limit. Routine operation is therefore limited by acidic variants, whose risk Section 7.2 accepts and Section 10 mitigates.

6 Design space

The design space of this step is the region of the four response-surface parameters over which every modelled attribute meets its intermediate acceptance criterion. It is judged on the 95 % prediction interval of a single run at manufacturing scale, as PCMP-001 §6.2 requires. A setting lies inside it when both ends of that interval lie within the criterion of every attribute. Osmolality and the four univariate parameters are held at their set-points, so the design space is conditioned on them, and it is not claimed beyond the ranges studied.

On the grid of 14,641 points, 73.7 % of the region studied lies inside the design space. Its boundary is set by the lower limit of acidic variants, which excludes 26.0 % of the grid points. Galactosylation excludes 0.3 % and afucosylation 0.1 %, and high mannose and aggregate exclude none. On the mean prediction in place of the single-run interval, 91.7 % of the grid would lie inside. A region drawn on mean predictions gives a single batch at its edge only about an even chance of meeting the criterion (PCMP-001 §6.2).

Figure 6.1 shows two planes of the design space in culture pH and pCO2. In the first, culture temperature and culture duration are at their set-points, and 3.1 % of the NOR of culture pH and pCO2, at its corner of high pH and high pCO2, lies outside the design space. In the second, culture temperature is at its NOR low and culture duration at its NOR high, the settings least favourable for acidic variants, and 18.8 % lies outside. The NOR therefore does not lie entirely inside the design space, and Section 7.2 gives the decision not to narrow it (PD-003-08).

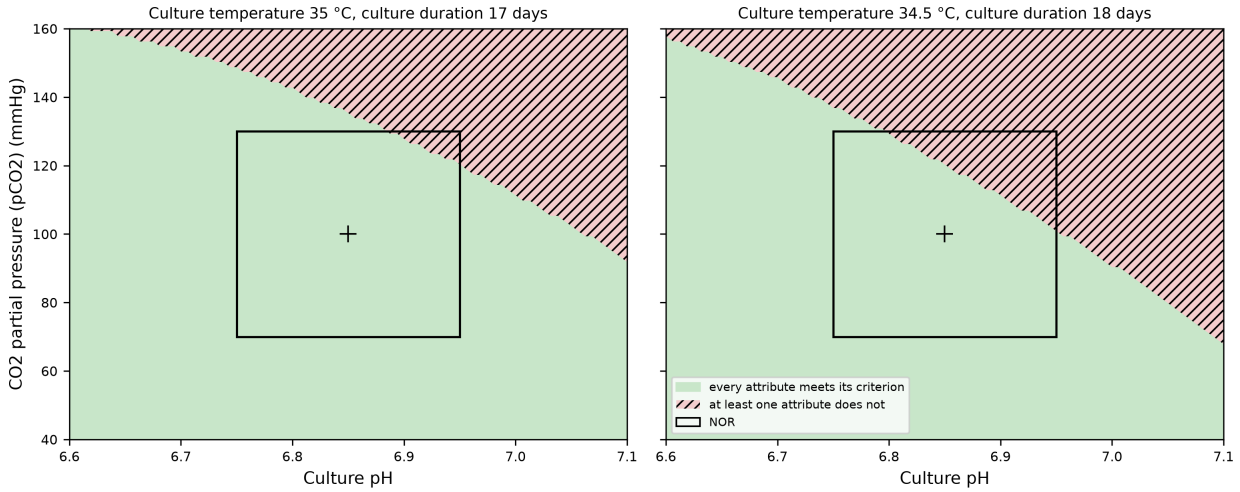


Figure 6.1: Two planes of the design space in culture pH and pCO2, on the 95 % prediction interval of a single run at manufacturing scale. Left, culture temperature and culture duration at their set-points. Right, culture temperature at its NOR low and culture duration at its NOR high. Plain green, every attribute meets its criterion. Hatched red, at least one does not. The rectangle is the NOR and the cross the set-point. The shares outside the design space given in the text come from a grid of 41 by 41 settings over the NOR of culture pH and pCO2.

Each PAR of Section 7.3 holds while the other parameters vary about their set-points, and the design space holds at every combination of settings at once. As ICH Q8(R2) states, working within the design space is not considered a change from a regulatory filing perspective (International Council for Harmonisation 2009). Movement within it is still assessed under the change management of the quality system, and a move outside it is a change (International Council for Harmonisation 2008). The design space rests on the scale-down model and is not confirmed at commercial scale at its edges.

7 Proven acceptable ranges

7.1 Acceptance criteria

Each output is judged against its intermediate acceptance criterion (Table 7.1). PCMP-001 §7.3 derives the criterion by back-calculation from the drug substance criterion, through the worst case of every downstream step on the platform, with no margin. Afucosylation, galactosylation and high mannose keep their drug substance criteria, because no downstream step changes them. The upper limit of acidic variants is the drug substance limit less the platform worst case of the increase across low-pH viral inactivation, 1.00 points, which gives 39.0 %. In the low-pH response-surface design every run at the same hold time gave the same acidic-variant result, so the full quadratic model reproduces the results exactly, cannot be judged against PCMP-001 §7.2, and the platform value stands. The increase applies to the upper limit only. The lower limit is the drug substance limit unchanged, which assumes that no downstream step lowers acidic variants (PCP-003 §12).

The aggregate criterion is the drug substance limit less the aggregate increase across UF/DF, multiplied by the aggregate clearance factor of cation exchange, less the aggregate increase across low-pH viral inactivation. PCMP-001 assumed that no step after cation exchange changes aggregate, and from the platform values alone the criterion is 9.30 %. Verification of the platform worst cases at manufacturing scale changed all three terms (PCMP-001 §7.5). UF/DF raised aggregate within its NOR in the univariate study S-UFDF-UNI2. The one-sided 95 % prediction bound of a single run, 0.10 points, replaced the assumption of no increase (PCR-010). The cation exchange model predicted within its NOR a lowest clearance of 1.58-fold at manufacturing scale, worse than the platform worst case of 2.00-fold. The one-sided 95 % prediction bound of a single batch there, 1.40-fold, replaced the platform value (PCR-007). These two bounds alone would set the criterion at 6.14 %. The low-pH viral inactivation model gave a bound of 0.54 points for the increase, better than the platform value of 0.70 points, on a model that meets the first three adequacy criteria of PCMP-001 §7.2 (PCR-006). That bound replaced the platform value too. The criterion in force is 6.30 %, stricter than the a priori value. It is computed from the unrounded terms, so the rounded values give 6.32 %. The increase in Table 7.1 is the sum of the two increases, and the UF/DF increase is carried back through the clearance factor of cation exchange.

PCMP-001 §7.5 asks for the risk of the change. The net change is a tightening. No in-process test of aggregate exists at this step, so no batch fails here, and the risk of the tighter criterion falls on the producer at release, where SEC tests aggregate in the drug substance against the limit of 5.0 % and detects an excursion. The widening alone moves the criterion 0.16 points in the direction of a higher risk to the patient. To

meet the drug substance limit, cation exchange must clear at least 1.40-fold, low-pH viral inactivation must add no more than 0.54 points and UF/DF no more than 0.10 points. At the highest single-run level that the model of this step predicts within the NOR, 2.35 % at manufacturing scale, the drug substance would then carry at most 2.17 %. None of the 2,000 batches of the integrated model fell outside the aggregate criterion of the drug substance, which puts the probability below 0.15 % at one-sided 95 % confidence (Section 8).

The widening rests on the bound of the low-pH viral inactivation model, from the 18 runs of its response-surface design. That model has predicted R^2 0.004 below R^2 and a lack-of-fit p-value of 0.504, and it also meets the criterion on residuals and influential runs. PCR-006 gives the bound and states whether it depends on a single run, as PCMP-001 §7.5 requires. The platform worst case is an increase over a hold of 180 min, while the NOR of the A-Mab hold as characterized ends at 120 min. At the edges of the range studied, which include the longer hold, A-Mab added up to 0.82 points, more than the platform value. The replacement therefore reflects the shorter hold of A-Mab and not a more stable product. It holds for routine manufacture, which is controlled within the hold-time NOR, and the NOR of hold time and temperature in force ends at the same longest hold and highest temperature as the NOR as characterized.

Without the model selection of INV-002 corrective action CA3, the verification would have used the full quadratic models, and the aggregate criterion would be 6.04 %. The low-pH model of acidic variants is an exact fit, which cannot replace the platform value, so the acidic-variant criterion would not change. Judged against that aggregate criterion, aggregate keeps its verdict over the NOR, its PARs and the qualification of its scale-down model.

Table 7.1: Intermediate acceptance criteria for the production bioreactor output and their derivation from the drug-substance criteria.

Attribute	DS criterion	Downstream worst case (platform or verified)		IAC	Basis
		Clearance factor	Increase		
Aggregates, HMW (%)	≤ 5.00	1.40	0.64	≤ 6.30	Carried back, tightened[a)] and widened[b)]
Acidic variants (%)	18.0–40.0	1.00	1.00	18.0–39.0	Carried back
Afucosylation (%)	2.00–13.0	1.00	0.00	2.00–13.0	DS criterion[c)]
Galactosylation (%)	10.0–40.0	1.00	0.00	10.0–40.0	DS criterion[c)]
High mannose (%)	3.00–10.0	1.00	0.00	3.00–10.0	DS criterion[c)]

[a)] The response-surface model of a downstream step predicted, within its NOR, a worse value than the platform worst case. The one-sided 95 % prediction bound of a single batch there replaced it.
[b)] The one-sided 95 % prediction bound of a single batch at a downstream step's A-Mab worst case within its NOR was better than the platform worst case. The bound replaced it.
[c)] No downstream step changes the attribute, so the drug-substance criterion applies.
DS, drug substance. IAC, intermediate acceptance criterion. Increase, in the unit of the criterion.
Computed from unrounded values. A criterion recomputed from the rounded values shown can differ in its last digit.

7.2 Outputs against their criteria over the NOR

Table 7.2 gives each output against its criterion, with the mean predicted at the set-point, its range over the NOR and the limits of the 95 % prediction interval of a single run over the NOR, which decide the verdict. Afucosylation, galactosylation, high mannose and aggregate meet their criteria. Acidic variants do not. At nine of the 81 combinations, all with pCO2 at its NOR high, the lower end of the prediction interval falls below 18.0 %. The least favourable combination is culture pH at its NOR high, culture temperature at its NOR low, culture duration at its NOR high and pCO2 at its NOR high. There the predicted median at manufacturing scale is 18.39 % and the prediction interval runs from 16.25 % to 20.82 %. The median lies 0.39 points above the lower limit, less than the bias of the model at the set-point (Section 5.3.1). With the initial viable cell concentration at its NOR low the median would fall to about 17.69 % (Section 5.4). The upper limit is met at every combination. The verdict is the same with the offset at either end of its interval (Section 3.1).

Table 7.2: Production bioreactor outputs against their intermediate acceptance criteria.

Attribute	Criterion	Mean predicted by the response-surface model[a]		95 % limits of a single run over the NOR[a]	Met across the NOR[b]
		At the set-point	Range over the NOR		
Afucosylation (%)	2.00–13.0	8.67	6.90 to 10.2	5.73 to 11.3	Yes
Galactosylation (%)	10.0–40.0	26.2	22.0 to 31.6	19.4 to 34.1	Yes
High mannose (%)	3.00–10.0	6.40[c]	5.75 to 7.03[c]	4.89 to 8.26[c]	Yes
Acidic variants (%)	18.0–39.0	22.8[c]	18.4 to 26.7[c]	16.2 to 30.3[c]	No
Aggregates, HMW (%)	≤ 6.30	1.64[c]	1.26 to 2.01[c]	1.09 to 2.35[c]	Yes

[a] At manufacturing scale: the scale-down prediction minus the offset of the scale-down model for this output, or divided by it where the offset is a ratio.
[b] Judged at every combination of NOR low, set-point and NOR high. Quality attributes: the 95 % prediction interval of a single run (t quantile, coefficient and residual uncertainty).
[c] Modelled on the log scale: the median, the back-transformed mean prediction.

PCP-003 §7 requires the NOR to lie inside the region over which every modelled attribute meets its criterion, and the NOR to be narrowed until it does. Every combination that fails has pCO2 at its NOR high, so the NOR of pCO2 is the one whose narrowing would act on all of them. Some fail with culture pH at its set-point, so narrowing the NOR of culture pH alone would not restore the criterion. The NOR was not narrowed (PD-003-08). It was kept, and the risk accepted as the producer's, on the following grounds.

1. Acidic variants have very low criticality, and their lower limit of 18.0 % was set from the range of results seen with the product.
2. On all 44 runs the NOR-propagated PARs contain the NOR, with a margin of 2.25 standard deviations of the routine variation of culture pH and 0.15 of pCO2 (Section 7.3). On the planned runs alone they do not (Section 5.3.4), so this ground holds only with the augmentation runs and the model selection.
3. A batch below the limit is detected by the release test of the drug substance against the acidic-variant criterion and does not reach the patient, so the risk of a failure falls on the producer.
4. In the integrated model of the train, one of the 2,000 simulated batches fell outside the acidic-variant criterion (Section 8). That model carries no effect of the initial viable cell concentration on acidic variants.

At the least favourable combination the fitted mean of the osmolality study lies below the lower limit at the NOR high of osmolality, although its slope is not significant

(Section 5.5). The same acceptance covers that setting, and it covers the initial viable cell concentration at its NOR low, where the univariate effect would take the median below the lower limit (PD-003-11). For the control strategy that PCMR-001 consolidates, this report recommends a combined alert in the control system, which fires when culture pH is above 6.79, pCO2 is above 101 mmHg and culture temperature is below 35.39 °C at the same time. It also recommends an action limit for acidic variants at harvest of 18.72 %, and the trending of acidic variants against the initial viable cell concentration of each production batch (Section 10). The release test of the drug substance detects a low result (ground 3).

7.3 PARs

A PAR is a range of a parameter within which the attribute is shown by supportive data to remain acceptable. ICH Q8(R2) defines it with the other parameters kept constant (International Council for Harmonisation 2009), which is the set-point analysis. The NOR-propagated analysis lets the other parameters vary across their NORs and judges the 95 % interval of the simulated runs, so it is stricter, and it is the PAR that supports a claim (PCMP-001 §6.4). Table 7.3 gives both for each of the four response-surface parameters and each of the five attributes, from the selected model of the attribute with the offset applied. Objective 3 requires a PAR where the selected model has a significant term in the parameter, 16 of the 20 combinations. The other four, culture duration for high mannose, pCO2 for high mannose, culture temperature for aggregate and pCO2 for aggregate, have no term in the parameter and are given for information.

Every NOR-propagated PAR contains the NOR, and 18 span the whole range studied. The exceptions are the PARs of culture pH and pCO2 for acidic variants. The PAR of culture pH ends 0.075 pH units above the NOR high, and the PAR of pCO2 ends 1.5 mmHg above it. Measured in standard deviations of the routine variation of each parameter, the width of its NOR divided by 6 (PCMP-001 §6.4), the margin is 2.25 for culture pH and 0.15 for pCO2. In the set-point analysis every PAR spans the range studied.

The range carried into the control strategy for each parameter is the intersection of its required PARs (PCP-003 §7, PCMP-001 §7.6). It is pH 6.6–7.02 for culture pH, 34–36 °C for culture temperature, 15–19 days for culture duration and 40–131 mmHg for pCO2.

Table 7.3: Proven acceptable ranges of the response-surface parameters for each attribute, at manufacturing scale.

Attribute	Criterion	Parameter	Unit	NOR	Required by objective 3	PAR, others at set-point	PAR, others across NOR
Afucosylation (%)	2–13	Culture pH	pH	6.75–6.95	yes	6.6–7.1	6.6–7.1
		Culture temperature	°C	34.5–35.5	yes	34–36	34–36
		Culture duration	days	16–18	yes	15–19	15–19
		CO2 partial pressure (pCO2)	mmHg	70–130	yes	40–160	40–160
Galactosylation (%)	10–40	Culture pH	pH	6.75–6.95	yes	6.6–7.1	6.6–7.1
		Culture temperature	°C	34.5–35.5	yes	34–36	34–36
		Culture duration	days	16–18	yes	15–19	15–19
		CO2 partial pressure (pCO2)	mmHg	70–130	yes	40–160	40–160
High mannose (%)	3–10	Culture pH	pH	6.75–6.95	yes	6.6–7.1	6.6–7.1
		Culture temperature	°C	34.5–35.5	yes	34–36	34–36
		Culture duration	days	16–18	no	15–19	15–19
		CO2 partial pressure (pCO2)	mmHg	70–130	no	40–160	40–160
Acidic variants (%)	18–39	Culture pH	pH	6.75–6.95	yes	6.6–7.1	6.6–7.02
		Culture temperature	°C	34.5–35.5	yes	34–36	34–36
		Culture duration	days	16–18	yes	15–19	15–19
		CO2 partial pressure (pCO2)	mmHg	70–130	yes	40–160	40–131
Aggregates, HMW (%)	≤ 6.3	Culture pH	pH	6.75–6.95	yes	6.6–7.1	6.6–7.1
		Culture temperature	°C	34.5–35.5	no	34–36	34–36
		Culture duration	days	16–18	yes	15–19	15–19
		CO2 partial pressure (pCO2)	mmHg	70–130	no	40–160	40–160

Every PAR is computed from the selected model of the attribute. PAR, others at set-point: the mean prediction with the other factors at their set-points. PAR, others across NOR: the 95 % interval of the simulated runs with the other factors drawn across their NORs, the PAR that supports a claim. Required by objective 3: the selected model has a significant term in the parameter. A PAR edge is rounded toward the inside of the range, to three significant figures. The range studied of each parameter is given in the table of process parameters.

The PAR of pCO2 is a range at the set-point of osmolality, because the response-surface model holds osmolality there. At manufacturing scale a higher pCO2 raises the osmolality as bicarbonate accumulates, and a higher osmolality lowered acidic variants in the screening design (Section 5.2). The PAR of pCO2 for acidic variants therefore does not include the effect of the osmolality rise that comes with it at scale.

Figure 7.1 to Figure 7.5 show, for each attribute, the PAR of the parameter with the largest main effect in its selected model. The acceptable region is shaded green, the NOR grey and the criterion dashed. The left panel is the set-point analysis and the right panel the NOR-propagated analysis, with the median and the 95 % interval of the simulated runs. Values are at manufacturing scale.

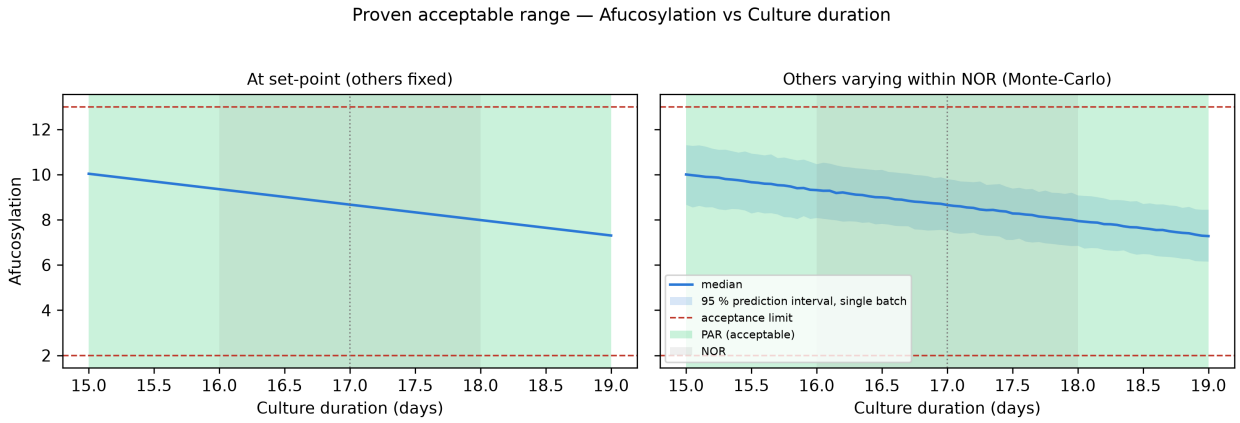


Figure 7.1: Proven acceptable range of afucosylation in culture duration. Left, the other factors at their set-points. Right, the other factors varying across their NORs.

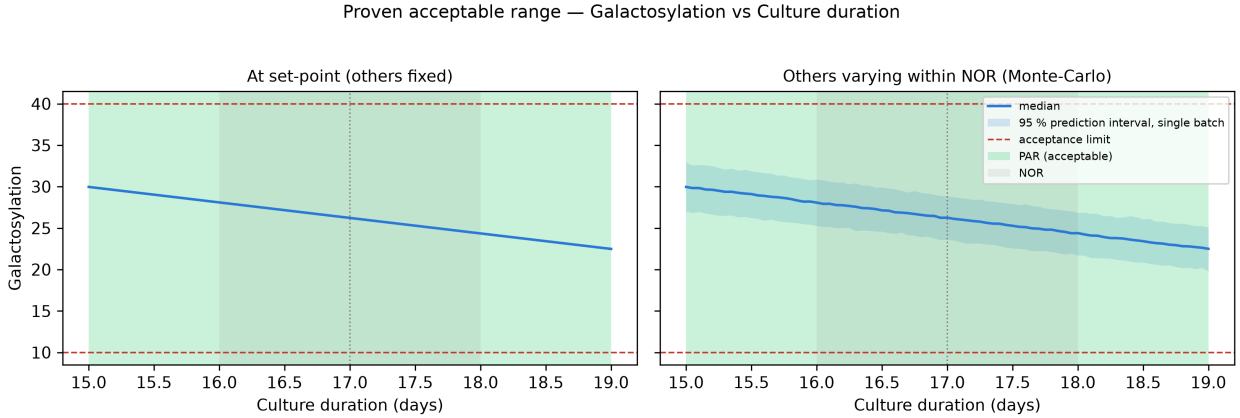


Figure 7.2: Proven acceptable range of galactosylation in culture duration. Left, the other factors at their set-points. Right, the other factors varying across their NORs.

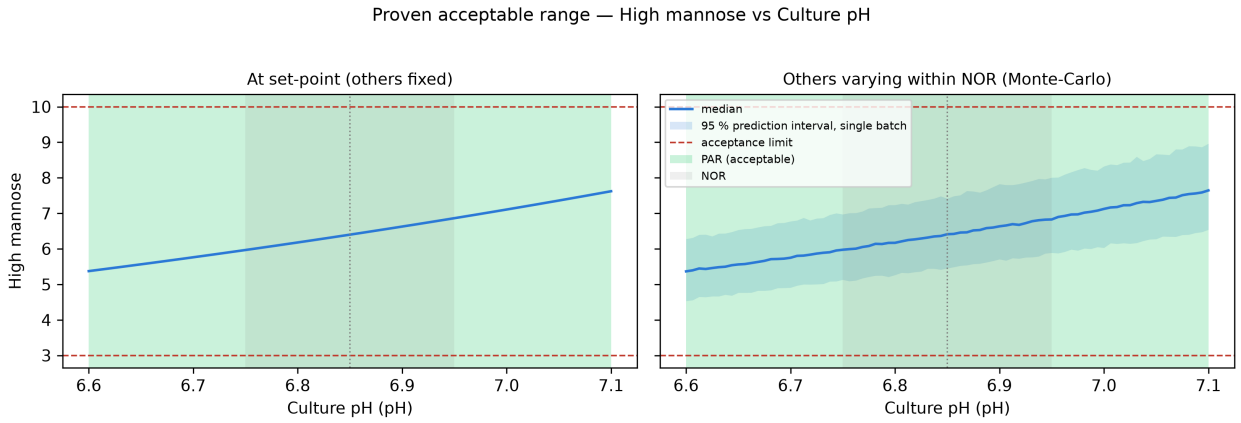


Figure 7.3: Proven acceptable range of high mannose in culture pH. Left, the other factors at their set-points. Right, the other factors varying across their NORs.

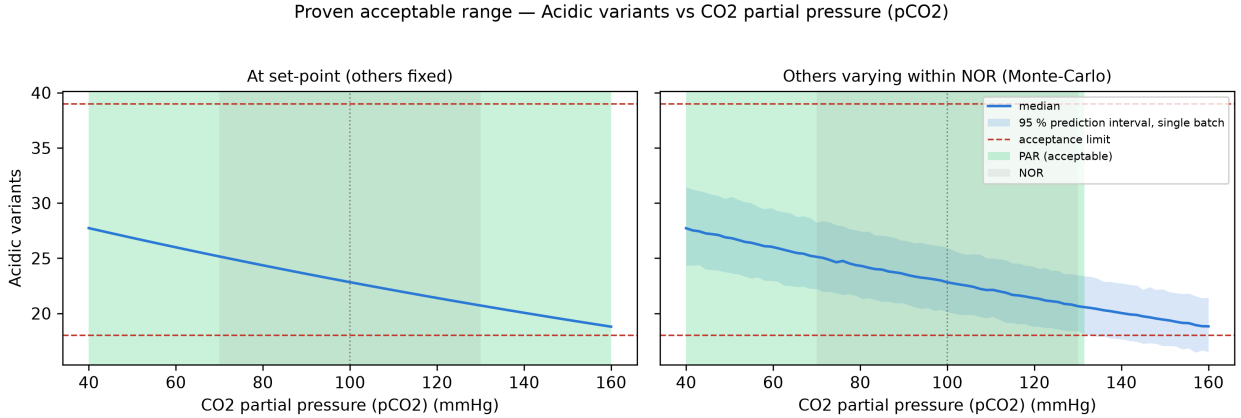


Figure 7.4: Proven acceptable range of acidic variants in pCO2. Left, the other factors at their set-points. Right, the other factors varying across their NORs. The NOR-propagated PAR ends just above the NOR high.

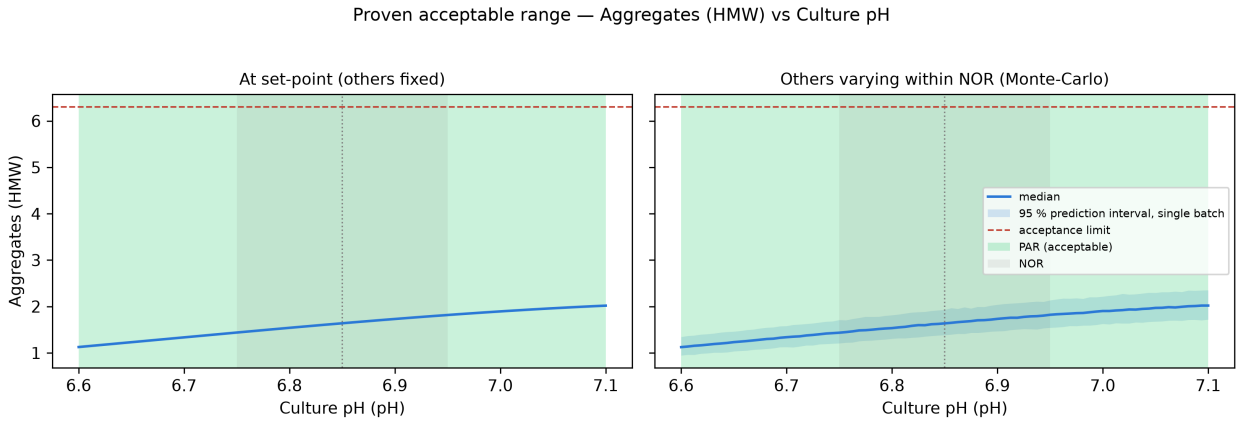


Figure 7.5: Proven acceptable range of aggregate in culture pH. Left, the other factors at their set-points. Right, the other factors varying across their NORs.

7.4 Univariate parameters and osmolality

The ranges of the four univariate parameters are supported by their own data, with the other parameters at their set-points (PCMP-001 §4). Every result met its criterion at both ends of each range (Table 5.10), so the range studied of each parameter is supported for the five attributes at that condition. For the initial viable cell concentration the support does not extend to the other parameters at their least favourable NOR settings for acidic variants (Section 5.4). Osmolality has no PAR from a model, because it was not a factor of the response-surface design. The osmolality study supports its NOR for every attribute except acidic variants, at the single combination of the other parameters that it tested, and no range of osmolality beyond its NOR is claimed.

8 Process capability and robustness

Table 8.1 gives the capability of the drug substance for the seven attributes that this step sets, from the 2,000 simulated batches of the integrated model. Each Cpk is judged against the target of 1.33 and the minimum of 1.00 of PCMP-001 §7.9. A Cpk between the two is acceptable, and a Cpk below the minimum needs an action in the control strategy.

The integrated model simulates manufacturing-scale batches through the whole train. It does not use the response-surface models of this report, and it applies no scale-down offset, because it simulates manufacturing scale directly. Its drug substance means differ from the predictions of Table 7.2, for afucosylation 7.98 % against 8.67 % predicted at the set-point. It draws every parameter of the step as the NOR-propagated analysis does, including osmolality, a higher level of which lowers acidic variants in its process model. It carries no effect of the initial viable cell concentration on acidic variants, so the capability does not include the effect found in the univariate study.

Table 8.1: Capability of the drug substance for the attributes that the production bioreactor sets, from 2,000 batches of the integrated model.

Attribute	Criti-cality	Criterion	Mean	SD	Cpk	Against the target and minimum	Batches outside	Normal tail (per million)
Afucosylation (%)	H	2–13	7.98	0.58	2.89	above the target	0	< 0.01
Galactosylation (%)	H	10–40	26.0	1.63	2.86	above the target	0	< 0.01
High mannose (%)	VH	3–10	5.99	0.51	1.94	above the target	0	< 0.01
Aggregates, HMW (%)	H	≤ 5	0.78	0.09	15.66	above the target	0	< 0.01
Acidic variants (%)	VL	18–40	22.4	1.39	1.05	between the minimum and the target	1	821
Host cell protein (ng/mg)	M-H	≤ 100	15.2	4.60	6.14	above the target	0	< 0.01
Residual DNA (ng/dose)	VL	≤ 0.001	1.7×10^{-4}	3.8×10^{-5}	7.30	above the target	0	< 0.01

Cpk is the lower of the two one-sided indices for a two-sided criterion. Batches outside, of 2,000. Normal tail, the tail probability of a normal distribution with the simulated mean and standard deviation. Residual DNA in ng per dose of 10 mg/kg for a 70 kg adult.

The Cpk of acidic variants is 1.05, between the minimum and the target, and it is limited by the lower limit, with an index of 1.05 for the lower limit and 4.24 for the upper. A higher pCO2, a higher culture pH and a lower culture temperature move acidic variants toward the lower limit (Section 5.3.3). In the simulation one of the 2,000 batches fell outside the criterion. PCMR-001 consolidates this capability with the acceptance of the acidic-variant risk (Section 7.2).

Every other attribute that the step sets has a Cpk above the target, and none of the simulated batches fell outside its criterion. With no simulated batch outside, the probability of a batch outside is below 0.15 % at one-sided 95 % confidence. That bound covers the sampling of the simulation and not the uncertainty of the integrated model. Aggregate in the drug substance lies far below its limit, at a mean of 0.78 %, and it is judged on the prediction interval of a single run over the NOR of this step (Section 7.2) and of the steps after it.

Host cell protein and residual DNA in the drug substance depend on the clearance of the downstream steps, which the integrated model takes from Protein A, cation exchange and anion exchange (PCR-005, PCR-007 and PCR-008). This step contributes the load, which Section 3.4 gives for information.

9 Parameter classification

The parameters were classified under SOP-4001 by the decision logic of PCMP-001 §7.7 (CMC Biotech Working Group 2009). A parameter whose variation affects a critical quality attribute within the range studied is a CPP when the risk that it falls outside its acceptable range in operation is not low. It is a WC-CPP when that risk is low, because the parameter is readily controlled and its PAR is wide against its NOR. A parameter that affects process performance and no critical quality attribute is a KPP. A parameter with no relevant effect on either is a GPP. Neither PCMP-001 nor SOP-4001 sets a numerical threshold for a relevant effect or a wide PAR, so each class below is argued from the effects found and from the margin between each PAR and its NOR.

- Culture pH is a WC-CPP. It changes all four critical attributes. Its NOR-propagated PARs for the critical attributes span the range studied, and its PAR for acidic variants ends 2.25 standard deviations of its routine variation above the NOR high. Base addition holds it at its set-point.
- Culture temperature is a WC-CPP. It changes afucosylation, galactosylation and high mannose, and its NOR-propagated PARs span the range studied for every attribute. The vessel jacket holds it.
- pCO2 is a WC-CPP. It changes galactosylation and afucosylation, and its NOR-propagated PARs for these attributes span the range studied. Its PAR for acidic variants, an attribute of very low criticality, ends 0.15 standard deviations of its routine variation above the NOR high. The carbon dioxide fraction of the sparge gas holds it.
- Osmolality is a CPP. It lowered galactosylation in the screening design and afucosylation in the osmolality study, both critical attributes. It has no PAR from a model, because it was not a factor of the response-surface design, so its PAR cannot be shown to be wide against its NOR. At its NOR high the fitted mean of acidic variants lies below their lower limit at the least favourable combination of the other parameters, although the slope of the osmolality study is not significant (Section 5.5). The risk of operating outside its acceptable range is therefore not shown to be low.
- Culture duration is a WC-CPP. It changes galactosylation, afucosylation and aggregate, and its NOR-propagated PARs span the range studied for every attribute. It is set by the day of harvest.
- Dissolved oxygen is a GPP. No effect was detected on any attribute measured, at the precision of Table 5.11. The culture is not limited by oxygen at either end of the range studied (Section 2.1), so no effect on growth or titre is expected either.
- The initial viable cell concentration is a KPP. It raised acidic variants, an attribute of very low criticality, by 3.51 points across its range, in a single significant result that may be a false positive (Section 5.4), and no effect on a critical attribute was

detected. A higher initial concentration brings the culture to its peak cell density earlier and raises the titre (Section 2.1), so it acts on process performance.
- Basal medium concentration is a KPP. No effect was detected on any attribute measured. A higher concentration raises the supply of nutrients, and with it growth and titre (PCP-003 §4.1), so it acts on process performance.
- Nutrient feed-1 volume is a KPP. No effect was detected on any attribute measured. The feed supplies the glucose, amino acids and precursors for growth and titre (Section 2.1), so it acts on process performance. The feed under-delivery of DEV-003-02 left the feed within its NOR (Section 13.2).

The classes of the three KPPs rest on their expected effect on growth and titre, which the record does not hold (PD-003-09). The step has four well-controlled critical process parameters, one CPP, three KPPs and one GPP.

10 Contribution to the control strategy

This step contributes the following elements to the control strategy of A-Mab. The combined alert, the action limit and the trending of items 2 to 4 are recommendations of this report, which PCMR-001 consolidates with those of the other steps.

1. The four WC-CPPs and the CPP osmolality are held within their NORs by the bioreactor control system. Culture pH is held by base addition, culture temperature by the vessel jacket, pCO2 by the carbon dioxide fraction of the sparge gas, osmolality through the base and feed additions, and culture duration by the day of harvest. The ranges carried into the control strategy are those of Section 7.3.
2. A combined alert in the control system, which fires when culture pH is above 6.79, pCO2 is above 101 mmHg and culture temperature is below 35.39 °C at the same time, the side of each that lowers acidic variants. A PAR is one-dimensional, so the PAR edge of one parameter holds only while the parameters that act with it stay within limits, and here those limits lie inside their NORs. Each limit is the most unfavourable value of its parameter at which the lower 95 % limit of a single run still meets the criterion of 18.0 %, with the other factors of the model at their unfavourable NOR edges. The alert covers every combination inside the NOR that fails the criterion. Osmolality is not a factor of the response-surface design, so the harvest action limit of item 3 covers it. The NOR is read as the range that the batch record controls, about ±3 standard deviations of the chance variation of each parameter (PCMP-001 §6.4).
3. An action limit for acidic variants at harvest of 18.72 %, the lower limit of the criterion plus 2 intermediate-precision standard deviations of icIEF (AMV-3013, 0.36 points). A result below it is found before purification, although it cannot change the batch.
4. Acidic variants at harvest of every production batch trended against the initial viable cell concentration of the batch, so that the univariate effect of Section 5.4 is confirmed or found to be a false positive.
5. The KPPs are held within their NORs for process consistency. The delivered feed is checked by the mass reconciliation after the run, which detected DEV-003-02. The basal medium is prepared at its set-point under SOP-2004.
6. The drug substance criteria of the attributes this step sets apply at release (PCMP-001 §3). The glycans, acidic variants and aggregate are tested there by the methods of Section 3.3.
7. The host cell protein that this step releases into the harvest is cleared by Protein A, cation exchange and anion exchange, and is judged at each of those steps and at the drug substance. Residual DNA is cleared by the same steps and judged at the drug substance only (PCR-005, PCR-007 and PCR-008).
8. Process performance qualification (PPQ) at the receiving site under PTP-001 runs the process at its set-points. It confirms the process at scale near the centre

of the NOR and does not test the edges of the ranges.

11 Discussion

The conclusions of this report rest on the scale-down model, whose offsets are known well enough that no verdict over the NOR depends on their exact size. Four facts limit how far that holds. The offset was estimated at the set-points only, so its use toward the edges of the range studied is an assumption. The carbon dioxide stripping of the commercial vessel was not matched, and osmolality was set independently of pCO2 (Section 3.2). The growth and titre of the runs are not in the record, so the model cannot be compared with manufacturing scale on process performance (PD-003-09). The difference in aggregate between the scales has no known cause (PD-003-13).

The following limitations apply to the conclusions of this report.

1. For aggregate only 11 of the 16 augmentation runs fall inside the 95 % prediction intervals of the model selected on the planned runs, so the single-run intervals of aggregate understate the scatter between runs (Section 5.3.4).
2. At the set-point the model of acidic variants predicts above the centre points and the manufacturing batches by more than the margin of its worst-corner median above the lower limit (Section 5.3.1). For acidic variants, 10 other models have an AICc within 2 units of the lowest, and only culture pH, culture temperature, pCO2 and culture pH × culture temperature are common to all of them. Each gives acidic variants the same verdict over the NOR.
3. Osmolality has no PAR from a model, and its interactions with the other parameters are known only from the screening design (PD-003-07).
4. The effect of the initial viable cell concentration on acidic variants comes from a single significant result of a univariate study, which cannot show its interactions, and the multivariate study that PCP-003 §6.4 calls for has not been made (PD-003-11). The capability does not include it.
5. The pCO2 settings of the augmentation and osmolality studies rest on equipment whose calibration status is not recorded (PD-003-12).

12 Conclusions

1. The relationship of each parameter with each attribute was quantified (objective 1). The selected response-surface models describe the five attributes over the ranges studied, and culture pH interacts with most of the other parameters.
2. The multivariate region was defined on the 95 % prediction interval of a single run (objective 2). It covers 73.7 % of the region studied, 59.6 % on the planned runs alone, and its boundary is set by the lower limit of acidic variants. The NOR does not lie entirely inside it (PD-003-08).
3. A PAR was established for each response-surface parameter and each attribute whose selected model has a significant term in it, 16 in all (objective 3). Every PAR contains the NOR on all runs, and on the planned runs alone three PARs for acidic variants do not. Osmolality has no PAR from a model.
4. Each parameter was classified (objective 4). The step has four WC-CPPs, one CPP, three KPPs and one GPP.
5. The capability supports the control strategy of Section 10 (objective 5). The lowest Cpk of the drug substance is that of acidic variants, 1.05, between the minimum and the target of PCMP-001, and every other attribute that the step sets has a Cpk above the target.
6. The scale-down model is qualified for every output with a criterion.
7. Against the completion criteria of PCP-003 §7, the study is complete except that the design space does not contain the NOR for acidic variants (PD-003-08), osmolality has no PAR from a model (PD-003-07), and the multivariate study of the initial viable cell concentration was not made (PD-003-11).

13 Study events and discrepancies from the plan

PCP-003 calls a departure from the plan a deviation. A process characterization study is not GMP manufacture, so this report records the events of execution as study events, under the numbers Version 1.0 gave them, and every departure from the plan as a discrepancy from the plan. A discrepancy is work that the plan did not foresee, or a requirement of the plan that was not met, however late it was found. Table 13.1 lists the two study events of this step, and Table 13.2 the 13 discrepancies from PCP-003.

Table 13.1: Study events recorded for the production bioreactor studies.

Event	Summary	Detected during	Outcome
DEV-003-01	pCO2 probe read high against the offline reference during screening run 4	in-process monitoring vs offline reference	run retained in the analysis
DEV-003-02	Nutrient feed-1 under-delivery on a centre-point batch	post-execution mass reconciliation	run retained in the analysis

13.1 DEV-003-01, pCO2 probe drift during screening run 4

During screening run 4 the pCO2 probe of the scale-down system, EQ-BRX-205, read 6.5 mmHg higher than the offline reference, and the comparison of the two in in-process monitoring detected the drift. The controller held the probe reading at the design level of 160 mmHg, so the run ran at about 153.5 mmHg. The record gives the cause as drift of the probe between two calibrations, and the equipment record shows the probe in calibration at the time of execution.

The run is analysed at its recorded setting, a shift of 0.108 in coded units, in which 1 is the half-range of pCO2. With the run at its design setting instead, the set of significant effects is the same for every attribute. With run 4 excluded the screening model keeps two residual degrees of freedom. Fewer effects then reach significance for galactosylation, high mannose, acidic variants and aggregate, and for afucosylation the same number, but not the same set. The run stayed in the analysis. The record names screening run 4 only. The response-surface design ran later on the same probe, and the record shows no recalibration between the two designs, so it does not show whether the probe read high in those runs too.

13.2 DEV-003-02, nutrient feed-1 under-delivery on a centre-point batch

A centre-point batch of the screening design received 4.2 % less nutrient feed-1 than its target, from lot LOT-FED-3120. The mass reconciliation after the run detected it, and its cause was a calibration offset of the feed pump. The batch received a feed volume of about 11.5 % of working volume, inside the NOR of 10–14 % of working volume and the range studied of 9–15 % of working volume, and the univariate study detected no effect of the nutrient feed-1 volume on any attribute across that range (Section 5.4). The record shows three determinations of the glycans of the batch by the glycan map (SOP-3010, validated under AMV-3010). Their mean has a 95 % confidence interval of ± 6.21 % of the result. The record names the batch as a screening centre point but not its run number, and Table 5.1 gives the spread of the three centre points. The run stayed in the analysis.

Table 13.2: Discrepancies from the plan PCP-003.

Discrepancy	Subject	Planned	Done	Reason
PD-003-01	Osmolality study at the least favourable NOR combination for acidic variants	PCP-003: osmolality in the screening design only, held at its set-point in the response-surface design	osmolality at its NOR low, set-point and NOR high, with the four response-surface parameters at their least favourable NOR combination for acidic variants, osmolality adjusted independently of pCO2 (study S-BRX-OSM), the number of runs from the precision of the prediction at the NOR high	the screening showed osmolality lowering acidic variants, the attribute nearest its lower limit, and the response-surface design could not show its size at the worst case; narrowing the osmolality NOR was not accepted, so the effect was measured

Discrepancy	Subject	Planned	Done	Reason
PD-003-02	Augmentation of the response-surface design	PCP-003: the face-centred central composite design of 28 runs in study S-BRX-RSM	16 further runs in a D-optimal augmentation, study S-BRX-AUG, as a second block	the full quadratic models of high mannose and acidic variants did not meet the limit on overfitting of PCMP-001 §7.2, and INV-002 calculated the number of runs
PD-003-03	Model selection for the response-surface outputs	PCP-003: the full quadratic model of each output (PCMP-001 §6 and §7.2)	for each output the scale chosen by the Box-Cox profile (the log scale where its 95 % interval contains 0 and excludes 1), the model with the lowest AICc among all hierarchical submodels of the full quadratic, a block term kept where the design has a second block, and that model judged against PCMP-001 §7.2; a scale-down offset taken on the scale of the model	the full quadratic carries terms the data do not support; selecting the model is standard practice, and INV-002 corrective action CA3 applies it to every report

Discrepancy	Subject	Planned	Done	Reason
PD-003-04	Conclusions resting on models that do not meet the criterion on residuals and influential runs	PCP-003: response-surface models meeting every model-adequacy criterion of PCMP-001 §7.2, used for the ranges and the design space, and each run with a Cook's distance above 4/n examined by deleting it and repeating the judgement	the models of the outputs that do not meet the criterion on residual patterns and influential runs (a residual trend at the level of the residual tests, or a run with a Cook's distance above the median of F(p, n − p)) used for the ranges and the design space, with the size of each trend stated against the criterion of its output and each run above the limit deleted in turn, the model selection repeated and the output judged over the NOR again	selected models of this step do not meet the criterion of PCMP-001 §7.2 on residual patterns and influential runs; whether a conclusion depends on the trend or on the run is shown by the size of the trend and by the deletion
PD-003-05	Univariate runs in duplicate	PCP-003: each univariate parameter run at the low and the high edge of its range (PCP-003 §6.4)	two runs at each edge	to measure the run-to-run variation within the study, against which a univariate effect is judged

Discrepancy	Subject	Planned	Done	Reason
PD-003-06	How osmolality was set in the screening design	PCP-003: the plan does not state how osmolality is set	osmolality set with sodium chloride added with each feed, pCO2 held at its level by the sparge, as in the later osmolality study	a design factor must be set by a stated procedure; the report states it
PD-003-07	Osmolality not carried into the response-surface design	PCP-003: a factor active in the screening design carried into the response-surface design, or the design augmented before the operating region is declared	osmolality, active on acidic variants and galactosylation in the screening design, held at its set-point in the response-surface design; its effect was measured afterwards in the osmolality study S-BRX-OSM (PD-003-01)	its range was judged too narrow to matter, against the screening rule of the plan

Discrepancy	Subject	Planned	Done	Reason
PD-003-08	Normal operating range not narrowed for acidic variants	PCP-003: where a NOR does not fit inside the operating region, the region reported as found and the NOR narrowed until it fits inside	the NOR kept, although at its least favourable combination the 95 % prediction interval of a single batch of acidic variants extends below their lower limit; acidic variants are monitored at harvest, and a combined alert fires when culture pH, pCO2 and culture temperature sit beyond conditional limits on the side that lowers acidic variants at the same time	the risk was accepted as the producer’s: acidic variants have very low criticality, release testing detects a low result, and the probability of an out-of-specification drug substance is quantified
PD-003-09	Viable cell density, viability and titre not in the study record	PCP-003: the viable cell density profile, the viability at harvest and the titre compared in the qualification and monitored through every run	none of the three is in the record of the qualification and characteri-zation runs, and they cannot be recovered	the monitoring records were kept in the scale-down system and not transferred to the study record

Discrepancy	Subject	Planned	Done	Reason
PD-003-10	Univariate study started before the response-surface study	PCP-003: the phases in the order of PCP-003 §11, the univariate assessments after the response-surface design	the univariate study S-BRX-UNI started before the response-surface study S-BRX-RSM	scheduling of the scale-down bioreactor system
PD-003-11	Initial viable cell concen-tration not carried into a multivariate study	PCP-003: a univariate parameter with an effect on a quality attribute that the mechanism does not account for carried into a multivariate study, and the plan amended	initial viable cell concentration, which acted on acidic variants in the univariate study, kept as a univariate parameter; this report sizes its effect against the criterion over the NOR, the risk acceptance of PD-003-08 is extended to its NOR low, and acidic variants at harvest are to be trended against it in production, because the effect is a single significant result among the univariate tests and may be a false positive	the univariate results were not analysed in Version 1.0

Discrepancy	Subject	Planned	Done	Reason
PD-003-12	Equipment and calibration status not recorded for the osmolality and augmentation studies	PCP-003: each study record states its execution window and the calibration status of the pCO2 probe it used (INV-001 preventive action PA3)	the records of the osmolality study S-BRX-OSM and of the augmentation runs name no equipment and state no calibration status, and they cannot be completed	the equipment was not entered in the study record
PD-003-13	Aggregate offset of the scale-down model accounted for, not investigated	PCP-003: Version 1.0: no statistically significant difference between the scales at $p < 0.05$, and the source of any difference investigated before any characterization run	aggregate differed between the scales at that level; no investigation was made before the characterization runs, and aggregate is judged through its offset under PCMP-001 Version 2.0 §7.1	a scale-down model may differ in mean from manufacturing scale where its offset is known precisely enough, the rule of PCMP-001 Version 2.0 §7.1, so the Version 1.0 test was stricter than the use of the model needs

13.3 PD-003-01, osmolality study

The osmolality study (Section 4.5) found a significant fall of afucosylation with osmolality, and no significant effect on acidic variants at the least favourable NOR combination of the other parameters (Section 5.5). Its fitted mean of acidic variants at the NOR high lies below the lower limit, and the acceptance of Section 7.2 covers that setting.

13.4 PD-003-02, augmentation of the response-surface design

The 16 runs added under INV-002 changed the selected models of four attributes, and the models selected on the planned runs predict them poorly for high mannose and aggregate. The PARs and the design space on the planned runs alone are given in Section 5.3.4. The full quadratic model of high mannose still exceeds the limit on overfitting on all runs.

13.5 PD-003-03, model selection

With the plan's full quadratic model in place of the selected models, on all 44 runs, high mannose would have no adequate model, and the PAR of pCO2 for acidic variants would not contain the NOR (Section 5.3.4). The selection also changes the criteria of this step through the verification of the downstream steps (Section 7.1).

13.6 PD-003-04, models that do not meet the criterion on residuals and influential runs

The selected models of galactosylation and acidic variants were used although their absolute residuals rise with the predicted value. No run of any model has a Cook's distance above the F median. With the standard deviation of the half of the predicted values in which each prediction lies, both attributes keep their verdict over the NOR (Section 5.3.2).

13.7 PD-003-05, duplicate univariate runs

The duplicate runs give the pooled standard deviation of the study, with 8 degrees of freedom, against which each univariate difference was tested, including the effect of the initial viable cell concentration on acidic variants (Section 5.4).

13.8 PD-003-06, setting of osmolality

Because osmolality was set independently of pCO2, the screening design and the osmolality study estimate the two effects separately, and the PAR of pCO2 does not include the rise of osmolality that comes with pCO2 at scale (Section 7.3).

13.9 PD-003-07, osmolality not carried into the response-surface design

The screening showed osmolality active on acidic variants and on galactosylation (Section 5.2), and the Version 2.0 re-analysis of the screening results found that it had not been carried forward. Every prediction and PAR and the design space of this report are therefore conditioned on osmolality at its set-point. The osmolality study measured its effect at one combination of the other parameters (PD-003-01), and its interactions with them are known from the screening design only.

13.10 PD-003-08, NOR not narrowed for acidic variants

Acidic variants do not meet their criterion over the NOR on the prediction interval of a single run (Section 7.2), and the NOR was not narrowed as PCP-003 §7 requires. The Version 2.0 re-analysis against the criteria of PCMP-001 Version 2.0 found the departure. The risk was accepted as the producer's, on the grounds that Section 7.2 gives. A combined alert in the control system covers every failing combination inside the NOR. It fires when culture pH is above 6.79, pCO2 is above 101 mmHg and culture temperature is below 35.39 °C at the same time. The action limit for acidic variants at harvest of 18.72 % and the trending of Section 10 complete the mitigation.

13.11 PD-003-09, viable cell density, viability and titre not in the record

PCP-003 §5.4 requires the culture to be monitored for viable cell density and viability, and §5.1 lists both, with the titre, among the outputs of the qualification. The monitoring records cannot be recovered. No criterion of this report depends on these outputs. Their absence means that the classes of the KPPs rest on platform knowledge for process performance (Section 9), and that the scale-down model was not compared with manufacturing scale on growth and titre.

13.12 PD-003-10, univariate study started before the response-surface study

The univariate study started on 2025-10-06, before the response-surface study started on 2025-10-20, and the review of the execution dates under INV-001 found it. The sequence of PCP-003 §11 exists so that the factors of the response-surface design are chosen from the screening result. The screening study had ended on 2025-10-03, before either study started, and the univariate parameters were assigned by RA-001 and were not candidates for the response-surface design. The order of the two studies therefore affects no analysis.

13.13 PD-003-11, initial viable cell concentration not carried into a multivariate study

The univariate study showed acidic variants 3.51 points higher at the high end of the initial viable cell concentration than at the low end, at manufacturing scale (Section 5.4). The mechanism of PCP-003 §4.3 does not account for the effect, so PCP-003 §6.4 carries the parameter into a multivariate study and amends the plan. That was not done, because the univariate results were not analysed in Version 1.0. The Version 2.0 re-analysis of the univariate results found the departure. With the parameter at its NOR low and the other parameters at their least favourable NOR combination, the effect would take the predicted median of acidic variants to about 17.69 %, below the lower limit. The effect is the only significant result among 28 univariate tests and may be a false positive. The risk acceptance of PD-003-08 is extended to that combination, and acidic variants at harvest are to be trended against the initial viable cell concentration of each production batch (Section 10). The verdict over the NOR and the capability of this report do not include the effect.

13.14 PD-003-12, equipment and calibration status not recorded for the osmolality and augmentation studies

The records of the osmolality study S-BRX-OSM and of the augmentation S-BRX-AUG name no equipment and state no calibration status of the pCO2 probe, which PCP-003 §9 and preventive action PA3 of INV-001 require. The review of the study record found the gap, and the records cannot be completed. If the studies used EQ-BRX-205, its next calibration was due on 2026-06-14, before both studies started. pCO2 varied from 40 mmHg to 160 mmHg in the 16 augmentation runs and was held at 130 mmHg in the osmolality study, so the accuracy of those settings is not documented. A probe that read high, as in study event DEV-003-01, would put the true pCO2 below the recorded level. Every selected model is fitted to the augmentation runs.

13.15 PD-003-13, aggregate offset of the scale-down model accounted for, not investigated

PCP-003 Version 1.0 qualified the scale-down model for an output only where the two scales showed no statistically significant difference at $p < 0.05$, and it required the source of any difference to be investigated before the characterization runs. For aggregate the two scales differ under that test ($p = 0.016$), and no investigation was made before the screening study started. The re-analysis of the qualification against the Version 1.0 test found it. Under PCMP-001 Version 2.0 §7.1 a scale-down model may differ in mean from manufacturing scale where its offset is known precisely enough, so the Version 1.0 test was stricter than the use of the model needs.

Aggregate is judged through its offset, a ratio of 0.917 with a 90 % confidence interval of 0.873 to 0.963, and the model is qualified for it (Table 3.1). The cause of the difference is not known.

Appendix A. Screening design matrix and responses

Table A.1 gives the settings and the results of the 19 runs of the screening study S-BRX-SCR, as measured in the scale-down model. The Run column of the design tables gives the standard order of each design, not the order in which the runs were made (Section 3.4). Run 4 is shown at its design setting of pCO2, and the analysis uses its recorded setting (Section 13.1).

Table A.1: Production bioreactor screening design, settings and results.

Run	Type	Factor settings					Responses				
		A	B	C	D	E	Afuc.	Gal.	High man-nose	Acidic	HMW
		pH	°C	mmHg	mOsm/kg	days	%	%	%	%	%
1		6.6	34	40	360	19	5.18	27.9	6.61	26.4	1.30
2		6.6	34	40	440	15	8.93	36.1	6.44	26.5	0.78
3		6.6	34	160	360	15	11.8	34.6	5.63	22.3	0.79
4[a)]		6.6	34	160	440	19	4.83	14.7	5.46	17.9	1.48
5		6.6	36	40	360	15	9.72	41.4	4.61	27.0	0.86
6		6.6	36	40	440	19	4.68	27.3	4.68	27.6	1.26
7		6.6	36	160	360	19	6.22	14.5	4.65	23.1	1.48
8		6.6	36	160	440	15	9.83	31.2	4.45	19.3	0.95
9	factorial	7.1	34	40	360	15	9.61	28.0	7.77	24.1	1.51
10		7.1	34	40	440	19	7.51	17.0	7.89	20.4	2.26
11		7.1	34	160	360	19	7.13	19.5	7.34	12.6	2.53
12		7.1	34	160	440	15	5.29	17.6	7.01	11.3	1.61
13		7.1	36	40	360	19	6.74	28.3	6.43	29.6	2.18
14		7.1	36	40	440	15	5.40	26.7	5.94	24.4	1.62
15		7.1	36	160	360	15	3.82	25.2	6.10	18.4	1.80
16		7.1	36	160	440	19	3.08	29.1	6.15	15.8	2.05
17		6.85	35	100	400	17	7.80	26.4	5.88	21.9	1.64
18	center	6.85	35	100	400	17	7.51	27.4	5.84	20.0	1.63
19		6.85	35	100	400	17	8.48	27.0	7.11	21.8	1.50

[a)] DEV-003-01: pCO2 probe read high against the offline reference during screening run 4. Run retained, and analysed at the recorded CO2 partial pressure (pCO2) of 153.5 mmHg in Version 2.0.
Factors: A, culture pH. B, culture temperature. C, CO2 partial pressure (pCO2). D, osmolality. E, culture duration.

Appendix B. Response-surface design matrix and responses

Table B.1 gives the settings and the results of the 28 planned runs of the response-surface study S-BRX-RSM, and Table B.2 those of the 16 runs of the augmentation S-BRX-AUG, as measured in the scale-down model.

Table B.1: Production bioreactor response-surface design, settings and results.

Run	Type	Factor settings				Responses				
		A	B	E	C	Afuc.	Gal.	High man-nose	Acidic	HMW
		pH	°C	days	mmHg	%	%	%	%	%
1	factorial	6.6	34	15	40	9.43	38.8	5.44	29.8	0.83
2		6.6	34	15	160	11.8	29.0	5.55	17.6	0.79
3		6.6	34	19	40	4.50	25.9	5.65	25.1	1.33
4		6.6	34	19	160	5.38	16.6	5.38	20.0	1.37
5		6.6	36	15	40	10.4	40.5	4.23	25.6	0.80
6		6.6	36	15	160	10.1	30.7	4.17	19.8	0.71
7		6.6	36	19	40	5.12	27.1	4.63	30.3	1.24
8		6.6	36	19	160	6.13	17.4	4.64	21.8	1.41
9		7.1	34	15	40	8.92	22.0	6.79	22.9	1.50
10		7.1	34	15	160	5.89	18.7	8.81	13.6	1.53
11		7.1	34	19	40	8.05	18.7	6.11	19.3	1.99
12		7.1	34	19	160	5.17	18.7	7.34	12.3	2.15
13		7.1	36	15	40	5.41	28.7	6.97	28.5	1.45
14		7.1	36	15	160	2.84	29.1	7.36	18.5	1.70
15		7.1	36	19	40	6.20	25.4	5.95	26.5	2.08
16		7.1	36	19	160	4.19	27.2	6.80	17.2	2.01
17		6.6	35	17	100	7.82	26.1	4.59	24.2	1.15
18		7.1	35	17	100	6.07	23.7	7.72	19.3	1.82
19	axial	6.85	34	17	100	8.77	24.3	6.75	19.9	1.51
20		6.85	36	17	100	7.55	27.9	5.51	26.8	1.32
21		6.85	35	15	100	10.3	30.8	6.11	20.8	1.30
22		6.85	35	19	100	6.89	22.5	5.54	22.4	2.06
23		6.85	35	17	40	8.48	27.1	6.39	25.7	1.50
24		6.85	35	17	160	8.39	23.5	5.94	19.2	1.45
25	center	6.85	35	17	100	7.88	23.0	5.75	20.5	1.52
26		6.85	35	17	100	7.95	27.3	6.32	22.0	1.60
27		6.85	35	17	100	7.35	25.0	5.60	22.4	1.55
28		6.85	35	17	100	7.27	28.2	4.96	21.1	1.37

Factors: A, culture pH. B, culture temperature. E, culture duration. C, CO2 partial pressure (pCO2).

Table B.2: Production bioreactor augmentation design, settings and results.

Run	Type	Factor settings				Responses				
		A	B	E	C	Afuc.	Gal.	High man-nose	Acidic	HMW
		pH	°C	days	mmHg	%	%	%	%	%
29	augmentation	6.6	34	15	160	11.7	31.6	5.92	18.8	0.68
30		6.6	34	17	40	8.07	32.1	5.23	31.5	1.00
31		6.6	34	19	100	4.55	22.0	5.34	23.5	1.52
32		6.6	35	19	160	5.69	17.2	4.76	19.5	1.23
33		6.6	36	15	40	9.47	36.2	4.71	28.5	0.75
34		6.6	36	17	160	8.59	23.3	5.29	20.5	1.09
35		6.6	36	19	40	4.56	26.4	4.56	29.0	1.56
36		6.85	34	15	40	12.6	30.9	6.61	23.6	1.13
37		6.85	34	19	160	7.22	19.0	6.44	17.6	1.83
38		6.85	36	15	160	7.49	29.5	5.51	21.5	1.20
39		7.1	34	15	100	7.55	22.4	6.61	18.1	1.61
40		7.1	34	17	160	6.17	20.0	7.21	13.4	1.97
41		7.1	34	19	40	7.71	19.7	7.83	21.6	2.29
42		7.1	35	15	160	4.34	24.0	6.83	15.9	1.52
43		7.1	36	17	40	7.03	28.6	6.74	25.6	2.05
44		7.1	36	19	160	3.57	26.1	7.45	17.4	2.38

Factors: A, culture pH. B, culture temperature. E, culture duration. C, CO2 partial pressure (pCO2).

Appendix C. Full effect and coefficient tables

Table C.1 to Table C.5 give every effect of the screening models, in the factor codes of Table 4.2. Table C.6 to Table C.13 give the coefficients and the analysis of variance of the selected response-surface models of the attributes other than acidic variants, whose tables are in Section 5.3.3.

Table C.1: Effects of the screening model of afucosylation, as measured in the scale-down model.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
AE	2.4500	1.2250	0.2651	4.6202	0.019	*
E	−2.3662	−1.1831	0.2651	−4.4623	0.021	*
AC	−1.7692	−0.8846	0.2670	−3.3133	0.045	*
A	−1.5770	−0.7885	0.2651	−2.9739	0.059	
B	−1.3505	−0.6752	0.2651	−2.5468	0.084	
D	−1.3211	−0.6605	0.2651	−2.4913	0.088	
AB	−1.2728	−0.6364	0.2651	−2.4003	0.096	
C	−0.7197	−0.3598	0.2670	−1.3478	0.270	
BD	0.4446	0.2223	0.2651	0.8384	0.463	
BE	0.3545	0.1772	0.2651	0.6684	0.552	
AD	−0.1827	−0.0913	0.2651	−0.3445	0.753	
BC	−0.1812	−0.0906	0.2670	−0.3393	0.757	
CD	−0.1373	−0.0686	0.2670	−0.2571	0.814	
DE	0.0434	0.0217	0.2651	0.0818	0.940	
CE	0.0217	0.0109	0.2670	0.0407	0.970	

Table C.2: Effects of the screening model of galactosylation, as measured in the scale-down model.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
E	−7.8645	−3.9322	0.1980	−19.8560	< 0.001	***
AE	6.9517	3.4758	0.1980	17.5514	< 0.001	***
C	−5.8404	−2.9202	0.1994	−14.6435	< 0.001	***
A	−4.5069	−2.2535	0.1980	−11.3789	0.001	**
BD	3.7451	1.8725	0.1980	9.4554	0.003	**
AC	3.6483	1.8241	0.1994	9.1472	0.003	**
B	3.5929	1.7964	0.1980	9.0712	0.003	**

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
AB	3.2307	1.6153	0.1980	8.1567	0.004	**
D	−2.5027	−1.2514	0.1980	−6.3188	0.008	**
CD	2.1311	1.0656	0.1994	5.3432	0.013	*
DE	1.8770	0.9385	0.1980	4.7390	0.018	*
BE	1.5430	0.7715	0.1980	3.8956	0.030	*
AD	−0.1460	−0.0730	0.1980	−0.3686	0.737	
BC	−0.0842	−0.0421	0.1994	−0.2112	0.846	
CE	0.0721	0.0360	0.1994	0.1808	0.868	

Table C.3: Effects of the screening model of high mannose, as measured in the scale-down model.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	1.5190	0.7595	0.1550	4.9011	0.016	*
B	−1.3852	−0.6926	0.1550	−4.4694	0.021	*
C	−0.4523	−0.2261	0.1560	−1.4493	0.243	
BC	0.3757	0.1879	0.1560	1.2040	0.315	
E	0.1513	0.0756	0.1550	0.4880	0.659	
D	−0.1455	−0.0727	0.1550	−0.4693	0.671	
AE	0.0962	0.0481	0.1550	0.3105	0.777	
AC	0.0961	0.0481	0.1560	0.3081	0.778	
DE	−0.0787	−0.0393	0.1550	−0.2538	0.816	
CE	−0.0617	−0.0308	0.1560	−0.1976	0.856	
BE	0.0529	0.0264	0.1550	0.1706	0.875	
AB	0.0363	0.0181	0.1550	0.1171	0.914	
CD	−0.0296	−0.0148	0.1560	−0.0949	0.930	
AD	−0.0171	−0.0085	0.1550	−0.0551	0.960	
BD	0.0063	0.0031	0.1550	0.0203	0.985	

Table C.4: Effects of the screening model of acidic variants, as measured in the scale-down model.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
C	−8.2064	−4.1032	0.2389	−17.1777	< 0.001	***
A	−4.1145	−2.0572	0.2372	−8.6725	0.003	**
B	3.0168	1.5084	0.2372	6.3588	0.008	**
D	−2.6064	−1.3032	0.2372	−5.4938	0.012	*
AB	1.9332	0.9666	0.2372	4.0748	0.027	*
AC	−1.8747	−0.9374	0.2389	−3.9241	0.029	*
BE	1.7815	0.8907	0.2372	3.7550	0.033	*
AD	−0.6177	−0.3089	0.2372	−1.3021	0.284	
CE	−0.5460	−0.2730	0.2389	−1.1429	0.336	
CD	−0.5448	−0.2724	0.2389	−1.1404	0.337	

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
BC	0.2038	0.1019	0.2389	0.4267	0.698	
BD	−0.1635	−0.0818	0.2372	−0.3447	0.753	
AE	0.0742	0.0371	0.2372	0.1564	0.886	
E	−0.0282	−0.0141	0.2372	−0.0594	0.956	
DE	−0.0193	−0.0096	0.2372	−0.0406	0.970	

Table C.5: Effects of the screening model of aggregate, as measured in the scale-down model.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	0.8332	0.4166	0.0207	20.1527	< 0.001	***
E	0.5782	0.2891	0.0207	13.9860	< 0.001	***
BE	−0.1416	−0.0708	0.0207	−3.4253	0.042	*
C	0.1168	0.0584	0.0208	2.8065	0.067	
CD	−0.0734	−0.0367	0.0208	−1.7624	0.176	
AD	−0.0673	−0.0336	0.0207	−1.6275	0.202	
AB	−0.0587	−0.0293	0.0207	−1.4191	0.251	
BD	−0.0554	−0.0277	0.0207	−1.3401	0.273	
DE	−0.0547	−0.0274	0.0207	−1.3237	0.277	
D	−0.0541	−0.0271	0.0207	−1.3091	0.282	
AE	0.0409	0.0205	0.0207	0.9898	0.395	
BC	−0.0243	−0.0122	0.0208	−0.5838	0.600	
CE	0.0230	0.0115	0.0208	0.5533	0.619	
AC	−0.0074	−0.0037	0.0208	−0.1784	0.870	
B	−0.0074	−0.0037	0.0207	−0.1785	0.870	

Table C.6: Coefficients of the selected model of afucosylation, on the raw scale and in coded units.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	8.2887	0.1597	51.8921	< 0.001	***
A	−0.9714	0.0986	−9.8562	< 0.001	***
B	−0.6617	0.0972	−6.8087	< 0.001	***
E	−1.3655	0.1003	−13.6081	< 0.001	***
C	−0.4624	0.0972	−4.7579	< 0.001	***
AB	−0.5479	0.1057	−5.1842	< 0.001	***
AE	1.3508	0.1098	12.3023	< 0.001	***
AC	−0.9592	0.1056	−9.0807	< 0.001	***
BE	0.4513	0.1079	4.1837	< 0.001	***
A^2	−1.3704	0.1844	−7.4332	< 0.001	***
Block	0.1145	0.0877	1.3056	0.201	

Table C.7: Analysis of variance of the selected model of afucosylation, on the raw scale.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	206.1987	10	20.6199	69.2572	< 0.001
Residual	9.8250	33	0.2977		
Lack of fit	9.4478	30	0.3149	2.5042	0.246
Pure error	0.3773	3	0.1258		

Table C.8: Coefficients of the selected model of galactosylation, on the raw scale and in coded units.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	25.9843	0.1912	135.9335	< 0.001	***
A	−2.0623	0.2184	−9.4435	< 0.001	***
B	1.9149	0.2153	8.8930	< 0.001	***
E	−3.7399	0.2214	−16.8914	< 0.001	***
C	−2.3167	0.2153	−10.7589	< 0.001	***
AB	1.8754	0.2341	8.0127	< 0.001	***
AE	2.6129	0.2432	10.7431	< 0.001	***
AC	2.0217	0.2341	8.6377	< 0.001	***
Block	0.1311	0.1912	0.6859	0.497	

Table C.9: Analysis of variance of the selected model of galactosylation, on the raw scale.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	1238.7982	8	154.8498	105.7974	< 0.001
Residual	51.2276	35	1.4636		
Lack of fit	34.8578	32	1.0893	0.1996	0.994
Pure error	16.3697	3	5.4566		

Table C.10: Coefficients of the selected model of high mannose, on the log scale and in coded units.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	1.7818	0.0123	144.9552	< 0.001	***
A	0.1748	0.0141	12.4277	< 0.001	***
B	−0.0618	0.0139	−4.4519	< 0.001	***
AB	0.0334	0.0151	2.2171	0.033	*
Block	0.0121	0.0123	0.9884	0.329	

Table C.11: Analysis of variance of the selected model of high mannose, on the log scale.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	1.1168	4	0.2792	45.7486	< 0.001
Residual	0.2380	39	0.0061		
Lack of fit	0.2081	36	0.0058	0.5805	0.820
Pure error	0.0299	3	0.0100		

Table C.12: Coefficients of the selected model of aggregate, on the log scale and in coded units.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	0.4070	0.0212	19.2299	< 0.001	***
A	0.2916	0.0131	22.2523	< 0.001	***
E	0.2314	0.0133	17.3567	< 0.001	***
AE	−0.0669	0.0146	−4.5844	< 0.001	***
A^2	−0.0830	0.0245	−3.3901	0.002	**
Block	0.0103	0.0116	0.8848	0.382	

Table C.13: Analysis of variance of the selected model of aggregate, on the log scale.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	4.2375	5	0.8475	159.9210	< 0.001
Residual	0.2014	38	0.0053		
Lack of fit	0.1880	35	0.0054	1.2034	0.514
Pure error	0.0134	3	0.0045		

Appendix D. Analytical methods summary

Table D.1 gives the analytical methods of the studies with their precision and limit of quantitation, as the method records hold them.

Table D.1: Analytical methods of the production bioreactor studies, with their precision and limit of quantitation.

Method	Procedure	Validation	Attributes	Sample	Precision	LOQ
N-Glycan Map (2-AB HILIC-UPLC)	SOP-3010	AMV-3010	Afucosylation, galactosylation and high mannose	harvest	2.5 % (relative)	0.5 % glycan
Size-Variants (SEC-HPLC)	SOP-3011	AMV-3011	Aggregate (HMW)	harvest	0.04 points (absolute SD)	not in the record
Charge Variants (icIEF)	SOP-3013	AMV-3013	Acidic variants	harvest	2 % (relative)	1 % acidic
Host-Cell Protein ELISA	SOP-3012	AMV-3012	Host cell protein	clarified harvest	9.5 % (relative)	1 ng/mg
Residual DNA (qPCR)	SOP-3014	AMV-3014	Residual DNA	clarified harvest	15 % (relative)	1 pg/mg

Precision as the relative standard deviation of repeat results, or for SEC as an absolute standard deviation in percentage points of HMW. LOQ, limit of quantitation.

References

Box, George E. P., and David R. Cox. 1964. "An Analysis of Transformations." *Journal of the Royal Statistical Society, Series B (Methodological)* 26 (2): 211–52.

Burnham, Kenneth P., and David R. Anderson. 2002. *Model Selection and Multimodel Inference: A Practical Information-Theoretic Approach*. 2nd ed. New York: Springer.

CMC Biotech Working Group. 2009. "A-Mab: A Case Study in Bioprocess Development." CASSS.

Fedorov, Valerii V. 1972. *Theory of Optimal Experiments*. New York: Academic Press.

Hurvich, Clifford M., and Chih-Ling Tsai. 1989. "Regression and Time Series Model Selection in Small Samples." *Biometrika* 76 (2): 297–307.

International Council for Harmonisation. 2008. "ICH Q10: Pharmaceutical Quality System." ICH.

———. 2009. "ICH Q8(R2): Pharmaceutical Development." ICH.

———. 2012. "ICH Q11: Development and Manufacture of Drug Substances." ICH.

———. 2023. "ICH Q9(R1): Quality Risk Management." ICH.

Montgomery, Douglas C. 2017. *Design and Analysis of Experiments*. 9th ed. Hoboken, NJ: Wiley.

U.S. Food and Drug Administration. 2011. "Guidance for Industry: Process Validation — General Principles and Practices." FDA.